# Supplementary Material: NetShield

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## I. Evidence Classification and Scope

The formal Raspberry Pi campaign contains 12 sessions and 216 accepted arms. Timing is reconstructed from accepted formal hardware timestamps; the parameter sweep and desynchronization study are offline software analyses, and the scaling study is discrete-event conformance. None is a new Raspberry Pi experiment. This supplementary build does not modify the frozen formal raw evidence.

TABLE I
Evidence classes.

| Component | Evidence type | Permitted interpretation |
|---|---|---|
| Formal campaign | Hardware | Three-Pi measurements |
| Full-raw audit | Raw reconstruction | Accepted-arm traceability |
| Parameter sweep | Offline replay | Frozen-selector sensitivity |
| Desynchronization | Software sensitivity | Phase counterfactual |
| Timing reconstruction | Raw reconstruction | Accepted-hardware timestamps |
| Scaling study | Discrete-event conformance | Analytical boundary check |

## II. Frozen Action Table and I-QPG Thresholds

TABLE II
Frozen action lattice and $v = 1$ thresholds against H-NONE.

| Action | Payload (B) | Utility | Threshold $Q$ (B) |
|---|---|---|---|
| L-NONE | 0 | 0.758501 | – |
| L-ALERT | 256 | 0.825168 | 221.84 |
| L-SUMMARY | 4096 | 0.891835 | 30.14 |
| M-NONE | 0 | 0.766968 | – |
| M-ALERT | 256 | 0.833635 | 254.91 |
| M-SUMMARY | 8192 | 0.900302 | 16.10 |
| H-NONE | 0 | 0.768378 | – |
| H-ALERT | 256 | 0.835044 | 260.41 |
| H-SUMMARY | 16384 | 0.901711 | 8.14 |
| H-FULL | 131072 | 0.968378 | 1.53 |

## III. Theorem Derivations

For $m$ remotely represented probes and $k$ full-evidence probes, payload is $mS + k(B - S)$. Hence one-round feasibility is

$$kB + (m - k)S \leq C,$$

and, for $C \geq mS$,

$$k^*(C, m) = \min \left\{ m, \left\lfloor \frac{C - mS}{B - S} \right\rfloor \right\}.$$

With $\rho = C/(nB)$, $\sigma = S/B$, $r = m/n$, and $f = k/n$, the continuous frontier is $\sigma r + (1 - \sigma) f \leq \rho$. For synchronized fixed-full overload, conservation gives $Q_\Sigma(T) = T(nB - C)$. At mid capacity the frozen I-QPG state alternates between H-FULL and H-NONE because the post-full queue exceeds every positive-payload threshold and the zero-payload round drains the residual.

# IV. Complete V-Sweep Results

The $v = 1$ identity gate covers 36 accepted D cells and four deterministic checks per cell. Low-capacity cycle/action composition changes with $v$; no all-$v$ cycle-identity claim is made.

TABLE III: Complete frozen-selector $v$-sweep results (432 retained rows).

| Session | $v$ | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---|---|---|---|---|---|---|
| F12__S01 | 1 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S02 | 1 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S03 | 1 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S04 | 1 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S05 | 1 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S06 | 1 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S07 | 1 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S08 | 1 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S09 | 1 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S10 | 1 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S11 | 1 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S12 | 1 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S01 | 1 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S02 | 1 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S03 | 1 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S04 | 1 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S05 | 1 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S06 | 1 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S07 | 1 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S08 | 1 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S09 | 1 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S10 | 1 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S11 | 1 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S12 | 1 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S01 | 1 | high | 1.000000 | 35389440 | 0 | 1 |
| F12__S02 | 1 | high | 1.000000 | 35389440 | 0 | 1 |
| F12__S03 | 1 | high | 1.000000 | 35389440 | 0 | 1 |
| F12__S04 | 1 | high | 1.000000 | 35389440 | 0 | 1 |
| F12__S05 | 1 | high | 1.000000 | 35389440 | 0 | 1 |
| F12__S06 | 1 | high | 1.000000 | 35389440 | 0 | 1 |
| F12__S07 | 1 | high | 1.000000 | 35389440 | 0 | 1 |
| F12__S08 | 1 | high | 1.000000 | 35389440 | 0 | 1 |
| F12__S09 | 1 | high | 1.000000 | 35389440 | 0 | 1 |
| F12__S10 | 1 | high | 1.000000 | 35389440 | 0 | 1 |
| F12__S11 | 1 | high | 1.000000 | 35389440 | 0 | 1 |
| F12__S12 | 1 | high | 1.000000 | 35389440 | 0 | 1 |
| F12__S01 | 3 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S02 | 3 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S03 | 3 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S04 | 3 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S05 | 3 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S06 | 3 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S07 | 3 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S08 | 3 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S09 | 3 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S10 | 3 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S11 | 3 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S12 | 3 | low | 0.244444 | 9191424 | 87382 | 4 |
| F12__S01 | 3 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S02 | 3 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S03 | 3 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S04 | 3 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S05 | 3 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12__S06 | 3 | mid | 0.500000 | 17694720 | 43691 | 2 |

| Session | $v$ | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---|---|---|---|---|---|---|
| F12_S07 | 3 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S08 | 3 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S09 | 3 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S10 | 3 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S11 | 3 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S12 | 3 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S01 | 3 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S02 | 3 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S03 | 3 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S04 | 3 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S05 | 3 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S06 | 3 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S07 | 3 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S08 | 3 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S09 | 3 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S10 | 3 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S11 | 3 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S12 | 3 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S01 | 10 | low | 0.300000 | 10838016 | 87386 | 10 |
| F12_S02 | 10 | low | 0.300000 | 10838016 | 87386 | 10 |
| F12_S03 | 10 | low | 0.300000 | 10838016 | 87386 | 10 |
| F12_S04 | 10 | low | 0.300000 | 10838016 | 87386 | 10 |
| F12_S05 | 10 | low | 0.300000 | 10838016 | 87386 | 10 |
| F12_S06 | 10 | low | 0.300000 | 10838016 | 87386 | 10 |
| F12_S07 | 10 | low | 0.300000 | 10838016 | 87386 | 10 |
| F12_S08 | 10 | low | 0.300000 | 10838016 | 87386 | 10 |
| F12_S09 | 10 | low | 0.300000 | 10838016 | 87386 | 10 |
| F12_S10 | 10 | low | 0.300000 | 10838016 | 87386 | 10 |
| F12_S11 | 10 | low | 0.300000 | 10838016 | 87386 | 10 |
| F12_S12 | 10 | low | 0.300000 | 10838016 | 87386 | 10 |
| F12_S01 | 10 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S02 | 10 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S03 | 10 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S04 | 10 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S05 | 10 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S06 | 10 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S07 | 10 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S08 | 10 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S09 | 10 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S10 | 10 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S11 | 10 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S12 | 10 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S01 | 10 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S02 | 10 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S03 | 10 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S04 | 10 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S05 | 10 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S06 | 10 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S07 | 10 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S08 | 10 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S09 | 10 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S10 | 10 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S11 | 10 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S12 | 10 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S01 | 30 | low | 0.322222 | 11476992 | 87398 | 28 |
| F12_S02 | 30 | low | 0.322222 | 11476992 | 87398 | 28 |
| F12_S03 | 30 | low | 0.322222 | 11476992 | 87398 | 28 |
| F12_S04 | 30 | low | 0.322222 | 11476992 | 87398 | 28 |
| F12_S05 | 30 | low | 0.322222 | 11476992 | 87398 | 28 |
| F12_S06 | 30 | low | 0.322222 | 11476992 | 87398 | 28 |
| F12_S07 | 30 | low | 0.322222 | 11476992 | 87398 | 28 |
| F12_S08 | 30 | low | 0.322222 | 11476992 | 87398 | 28 |

| Session | $v$ | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---|---|---|---|---|---|---|
| F12_S09 | 30 | low | 0.322222 | 11476992 | 87398 | 28 |
| F12_S10 | 30 | low | 0.322222 | 11476992 | 87398 | 28 |
| F12_S11 | 30 | low | 0.322222 | 11476992 | 87398 | 28 |
| F12_S12 | 30 | low | 0.322222 | 11476992 | 87398 | 28 |
| F12_S01 | 30 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S02 | 30 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S03 | 30 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S04 | 30 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S05 | 30 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S06 | 30 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S07 | 30 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S08 | 30 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S09 | 30 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S10 | 30 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S11 | 30 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S12 | 30 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S01 | 30 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S02 | 30 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S03 | 30 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S04 | 30 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S05 | 30 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S06 | 30 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S07 | 30 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S08 | 30 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S09 | 30 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S10 | 30 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S11 | 30 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S12 | 30 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S01 | 100 | low | 0.333333 | 11821056 | 87436 | – |
| F12_S02 | 100 | low | 0.333333 | 11821056 | 87436 | – |
| F12_S03 | 100 | low | 0.333333 | 11821056 | 87436 | – |
| F12_S04 | 100 | low | 0.333333 | 11821056 | 87436 | – |
| F12_S05 | 100 | low | 0.333333 | 11821056 | 87436 | – |
| F12_S06 | 100 | low | 0.333333 | 11821056 | 87436 | – |
| F12_S07 | 100 | low | 0.333333 | 11821056 | 87436 | – |
| F12_S08 | 100 | low | 0.333333 | 11821056 | 87436 | – |
| F12_S09 | 100 | low | 0.333333 | 11821056 | 87436 | – |
| F12_S10 | 100 | low | 0.333333 | 11821056 | 87436 | – |
| F12_S11 | 100 | low | 0.333333 | 11821056 | 87436 | – |
| F12_S12 | 100 | low | 0.333333 | 11821056 | 87436 | – |
| F12_S01 | 100 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S02 | 100 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S03 | 100 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S04 | 100 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S05 | 100 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S06 | 100 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S07 | 100 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S08 | 100 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S09 | 100 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S10 | 100 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S11 | 100 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S12 | 100 | mid | 0.500000 | 17694720 | 43691 | 2 |
| F12_S01 | 100 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S02 | 100 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S03 | 100 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S04 | 100 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S05 | 100 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S06 | 100 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S07 | 100 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S08 | 100 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S09 | 100 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S10 | 100 | high | 1.000000 | 35389440 | 0 | 1 |

| Session | $v$ | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---|---|---|---|---|---|---|
| F12_S11 | 100 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S12 | 100 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S01 | 168 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S02 | 168 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S03 | 168 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S04 | 168 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S05 | 168 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S06 | 168 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S07 | 168 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S08 | 168 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S09 | 168 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S10 | 168 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S11 | 168 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S12 | 168 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S01 | 168 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S02 | 168 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S03 | 168 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S04 | 168 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S05 | 168 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S06 | 168 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S07 | 168 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S08 | 168 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S09 | 168 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S10 | 168 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S11 | 168 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S12 | 168 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S01 | 168 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S02 | 168 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S03 | 168 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S04 | 168 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S05 | 168 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S06 | 168 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S07 | 168 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S08 | 168 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S09 | 168 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S10 | 168 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S11 | 168 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S12 | 168 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S01 | 300 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S02 | 300 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S03 | 300 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S04 | 300 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S05 | 300 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S06 | 300 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S07 | 300 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S08 | 300 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S09 | 300 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S10 | 300 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S11 | 300 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S12 | 300 | low | 0.244444 | 9209088 | 87382 | 4 |
| F12_S01 | 300 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S02 | 300 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S03 | 300 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S04 | 300 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S05 | 300 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S06 | 300 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S07 | 300 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S08 | 300 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S09 | 300 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S10 | 300 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S11 | 300 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S12 | 300 | mid | 0.500000 | 17729280 | 43691 | 2 |

| Session | $v$ | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---|---|---|---|---|---|---|
| F12_S01 | 300 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S02 | 300 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S03 | 300 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S04 | 300 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S05 | 300 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S06 | 300 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S07 | 300 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S08 | 300 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S09 | 300 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S10 | 300 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S11 | 300 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S12 | 300 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S01 | 1000 | low | 0.288889 | 10582272 | 87896 | 7 |
| F12_S02 | 1000 | low | 0.288889 | 10582272 | 87896 | 7 |
| F12_S03 | 1000 | low | 0.288889 | 10582272 | 87896 | 7 |
| F12_S04 | 1000 | low | 0.288889 | 10582272 | 87896 | 7 |
| F12_S05 | 1000 | low | 0.288889 | 10582272 | 87896 | 7 |
| F12_S06 | 1000 | low | 0.288889 | 10582272 | 87896 | 7 |
| F12_S07 | 1000 | low | 0.288889 | 10582272 | 87896 | 7 |
| F12_S08 | 1000 | low | 0.288889 | 10582272 | 87896 | 7 |
| F12_S09 | 1000 | low | 0.288889 | 10582272 | 87896 | 7 |
| F12_S10 | 1000 | low | 0.288889 | 10582272 | 87896 | 7 |
| F12_S11 | 1000 | low | 0.288889 | 10582272 | 87896 | 7 |
| F12_S12 | 1000 | low | 0.288889 | 10582272 | 87896 | 7 |
| F12_S01 | 1000 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S02 | 1000 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S03 | 1000 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S04 | 1000 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S05 | 1000 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S06 | 1000 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S07 | 1000 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S08 | 1000 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S09 | 1000 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S10 | 1000 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S11 | 1000 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S12 | 1000 | mid | 0.500000 | 17729280 | 43691 | 2 |
| F12_S01 | 1000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S02 | 1000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S03 | 1000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S04 | 1000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S05 | 1000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S06 | 1000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S07 | 1000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S08 | 1000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S09 | 1000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S10 | 1000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S11 | 1000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S12 | 1000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S01 | 3000 | low | 0.244444 | 9491712 | 87382 | 4 |
| F12_S02 | 3000 | low | 0.244444 | 9491712 | 87382 | 4 |
| F12_S03 | 3000 | low | 0.244444 | 9491712 | 87382 | 4 |
| F12_S04 | 3000 | low | 0.244444 | 9491712 | 87382 | 4 |
| F12_S05 | 3000 | low | 0.244444 | 9491712 | 87382 | 4 |
| F12_S06 | 3000 | low | 0.244444 | 9491712 | 87382 | 4 |
| F12_S07 | 3000 | low | 0.244444 | 9491712 | 87382 | 4 |
| F12_S08 | 3000 | low | 0.244444 | 9491712 | 87382 | 4 |
| F12_S09 | 3000 | low | 0.244444 | 9491712 | 87382 | 4 |
| F12_S10 | 3000 | low | 0.244444 | 9491712 | 87382 | 4 |
| F12_S11 | 3000 | low | 0.244444 | 9491712 | 87382 | 4 |
| F12_S12 | 3000 | low | 0.244444 | 9491712 | 87382 | 4 |
| F12_S01 | 3000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S02 | 3000 | mid | 0.500000 | 18247680 | 43691 | 2 |

| Session | $v$ | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---|---|---|---|---|---|---|
| F12_S03 | 3000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S04 | 3000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S05 | 3000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S06 | 3000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S07 | 3000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S08 | 3000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S09 | 3000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S10 | 3000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S11 | 3000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S12 | 3000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S01 | 3000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S02 | 3000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S03 | 3000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S04 | 3000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S05 | 3000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S06 | 3000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S07 | 3000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S08 | 3000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S09 | 3000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S10 | 3000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S11 | 3000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S12 | 3000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S01 | 10000 | low | 0.244444 | 9756672 | 87382 | 4 |
| F12_S02 | 10000 | low | 0.244444 | 9756672 | 87382 | 4 |
| F12_S03 | 10000 | low | 0.244444 | 9756672 | 87382 | 4 |
| F12_S04 | 10000 | low | 0.244444 | 9756672 | 87382 | 4 |
| F12_S05 | 10000 | low | 0.244444 | 9756672 | 87382 | 4 |
| F12_S06 | 10000 | low | 0.244444 | 9756672 | 87382 | 4 |
| F12_S07 | 10000 | low | 0.244444 | 9756672 | 87382 | 4 |
| F12_S08 | 10000 | low | 0.244444 | 9756672 | 87382 | 4 |
| F12_S09 | 10000 | low | 0.244444 | 9756672 | 87382 | 4 |
| F12_S10 | 10000 | low | 0.244444 | 9756672 | 87382 | 4 |
| F12_S11 | 10000 | low | 0.244444 | 9756672 | 87382 | 4 |
| F12_S12 | 10000 | low | 0.244444 | 9756672 | 87382 | 4 |
| F12_S01 | 10000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S02 | 10000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S03 | 10000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S04 | 10000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S05 | 10000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S06 | 10000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S07 | 10000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S08 | 10000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S09 | 10000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S10 | 10000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S11 | 10000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S12 | 10000 | mid | 0.500000 | 18247680 | 43691 | 2 |
| F12_S01 | 10000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S02 | 10000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S03 | 10000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S04 | 10000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S05 | 10000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S06 | 10000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S07 | 10000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S08 | 10000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S09 | 10000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S10 | 10000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S11 | 10000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S12 | 10000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S01 | 30000 | low | 0.288889 | 11489280 | 99672 | 7 |
| F12_S02 | 30000 | low | 0.288889 | 11489280 | 99672 | 7 |
| F12_S03 | 30000 | low | 0.288889 | 11489280 | 99672 | 7 |
| F12_S04 | 30000 | low | 0.288889 | 11489280 | 99672 | 7 |

| Session | $v$ | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---|---|---|---|---|---|---|
| F12_S05 | 30000 | low | 0.288889 | 11489280 | 99672 | 7 |
| F12_S06 | 30000 | low | 0.288889 | 11489280 | 99672 | 7 |
| F12_S07 | 30000 | low | 0.288889 | 11489280 | 99672 | 7 |
| F12_S08 | 30000 | low | 0.288889 | 11489280 | 99672 | 7 |
| F12_S09 | 30000 | low | 0.288889 | 11489280 | 99672 | 7 |
| F12_S10 | 30000 | low | 0.288889 | 11489280 | 99672 | 7 |
| F12_S11 | 30000 | low | 0.288889 | 11489280 | 99672 | 7 |
| F12_S12 | 30000 | low | 0.288889 | 11489280 | 99672 | 7 |
| F12_S01 | 30000 | mid | 0.500000 | 18800640 | 43691 | 2 |
| F12_S02 | 30000 | mid | 0.500000 | 18800640 | 43691 | 2 |
| F12_S03 | 30000 | mid | 0.500000 | 18800640 | 43691 | 2 |
| F12_S04 | 30000 | mid | 0.500000 | 18800640 | 43691 | 2 |
| F12_S05 | 30000 | mid | 0.500000 | 18800640 | 43691 | 2 |
| F12_S06 | 30000 | mid | 0.500000 | 18800640 | 43691 | 2 |
| F12_S07 | 30000 | mid | 0.500000 | 18800640 | 43691 | 2 |
| F12_S08 | 30000 | mid | 0.500000 | 18800640 | 43691 | 2 |
| F12_S09 | 30000 | mid | 0.500000 | 18800640 | 43691 | 2 |
| F12_S10 | 30000 | mid | 0.500000 | 18800640 | 43691 | 2 |
| F12_S11 | 30000 | mid | 0.500000 | 18800640 | 43691 | 2 |
| F12_S12 | 30000 | mid | 0.500000 | 18800640 | 43691 | 2 |
| F12_S01 | 30000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S02 | 30000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S03 | 30000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S04 | 30000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S05 | 30000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S06 | 30000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S07 | 30000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S08 | 30000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S09 | 30000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S10 | 30000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S11 | 30000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S12 | 30000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S01 | 100000 | low | 0.288889 | 11796480 | 142052 | – |
| F12_S02 | 100000 | low | 0.288889 | 11796480 | 142052 | – |
| F12_S03 | 100000 | low | 0.288889 | 11796480 | 142052 | – |
| F12_S04 | 100000 | low | 0.288889 | 11796480 | 142052 | – |
| F12_S05 | 100000 | low | 0.288889 | 11796480 | 142052 | – |
| F12_S06 | 100000 | low | 0.288889 | 11796480 | 142052 | – |
| F12_S07 | 100000 | low | 0.288889 | 11796480 | 142052 | – |
| F12_S08 | 100000 | low | 0.288889 | 11796480 | 142052 | – |
| F12_S09 | 100000 | low | 0.288889 | 11796480 | 142052 | – |
| F12_S10 | 100000 | low | 0.288889 | 11796480 | 142052 | – |
| F12_S11 | 100000 | low | 0.288889 | 11796480 | 142052 | – |
| F12_S12 | 100000 | low | 0.288889 | 11796480 | 142052 | – |
| F12_S01 | 100000 | mid | 0.622222 | 22855680 | 95575 | 8 |
| F12_S02 | 100000 | mid | 0.622222 | 22855680 | 95575 | 8 |
| F12_S03 | 100000 | mid | 0.622222 | 22855680 | 95575 | 8 |
| F12_S04 | 100000 | mid | 0.622222 | 22855680 | 95575 | 8 |
| F12_S05 | 100000 | mid | 0.622222 | 22855680 | 95575 | 8 |
| F12_S06 | 100000 | mid | 0.622222 | 22855680 | 95575 | 8 |
| F12_S07 | 100000 | mid | 0.622222 | 22855680 | 95575 | 8 |
| F12_S08 | 100000 | mid | 0.622222 | 22855680 | 95575 | 8 |
| F12_S09 | 100000 | mid | 0.622222 | 22855680 | 95575 | 8 |
| F12_S10 | 100000 | mid | 0.622222 | 22855680 | 95575 | 8 |
| F12_S11 | 100000 | mid | 0.622222 | 22855680 | 95575 | 8 |
| F12_S12 | 100000 | mid | 0.622222 | 22855680 | 95575 | 8 |
| F12_S01 | 100000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S02 | 100000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S03 | 100000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S04 | 100000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S05 | 100000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S06 | 100000 | high | 1.000000 | 35389440 | 0 | 1 |

| Session | $v$ | Cap. | Core frac. | Payload (B) | Max Q (B) | Period |
|---|---|---|---|---|---|---|
| F12_S07 | 100000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S08 | 100000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S09 | 100000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S10 | 100000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S11 | 100000 | high | 1.000000 | 35389440 | 0 | 1 |
| F12_S12 | 100000 | high | 1.000000 | 35389440 | 0 | 1 |

## V. Complete Desynchronization Sensitivity

Probe phases are independently assigned from fixed seeds. Probes exchange neither action nor queue state, and no joint admission is introduced.

TABLE IV: Complete desynchronization software-sensitivity results (240 retained rows).

| Seed | Cap. | Method | Core frac. | On-time | Max Q (B) | Phases (ms) |
|---|---|---|---|---|---|---|
| 2026090901 | low | desynchronized_independent | 0.333333 | 0.000000 | 131029 | 595/193/691 |
| 2026090901 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090901 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090901 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090901 | mid | desynchronized_independent | 0.666667 | 0.505556 | 130985 | 595/193/691 |
| 2026090901 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090901 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090901 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090901 | high | desynchronized_independent | 1.000000 | 1.000000 | 121111 | 595/193/691 |
| 2026090901 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090901 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090901 | high | shielded_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090902 | low | desynchronized_independent | 0.333333 | 0.000000 | 131029 | 403/632/306 |
| 2026090902 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090902 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090902 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090902 | mid | desynchronized_independent | 0.666667 | 0.500000 | 130985 | 403/632/306 |
| 2026090902 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090902 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090902 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090902 | high | desynchronized_independent | 1.000000 | 1.000000 | 130876 | 403/632/306 |
| 2026090902 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090902 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090902 | high | shielded_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090903 | low | desynchronized_independent | 0.333333 | 0.000000 | 131029 | 489/690/996 |
| 2026090903 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090903 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090903 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090903 | mid | desynchronized_independent | 0.666667 | 0.500000 | 130985 | 489/690/996 |
| 2026090903 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090903 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090903 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090903 | high | desynchronized_independent | 1.000000 | 1.000000 | 1180 | 489/690/996 |
| 2026090903 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090903 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090903 | high | shielded_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090904 | low | desynchronized_independent | 0.333333 | 0.000000 | 131028 | 945/680/799 |
| 2026090904 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090904 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090904 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090904 | mid | desynchronized_independent | 0.666667 | 0.500000 | 130985 | 945/680/799 |
| 2026090904 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090904 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090904 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090904 | high | desynchronized_independent | 1.000000 | 1.000000 | 21234 | 945/680/799 |
| 2026090904 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090904 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090904 | high | shielded_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090905 | low | desynchronized_independent | 0.333333 | 0.000000 | 131028 | 154/172/828 |
| 2026090905 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090905 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090905 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090905 | mid | desynchronized_independent | 0.666667 | 0.505556 | 130985 | 154/172/828 |
| 2026090905 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090905 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |

| Seed | Cap. | Method | Core frac. | On-time | Max Q (B) | Phases (ms) |
|---|---|---|---|---|---|---|
| 2026090905 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090905 | high | desynchronized_independent | 1.000000 | 1.000000 | 130876 | 154/172/828 |
| 2026090905 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090905 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090905 | high | shielded_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090906 | low | desynchronized_independent | 0.333333 | 0.000000 | 131029 | 60/626/317 |
| 2026090906 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090906 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090906 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090906 | mid | desynchronized_independent | 0.666667 | 0.505556 | 130985 | 60/626/317 |
| 2026090906 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090906 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090906 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090906 | high | desynchronized_independent | 1.000000 | 1.000000 | 130876 | 60/626/317 |
| 2026090906 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090906 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090906 | high | shielded_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090907 | low | desynchronized_independent | 0.333333 | 0.000000 | 131029 | 753/955/576 |
| 2026090907 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090907 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090907 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090907 | mid | desynchronized_independent | 0.666667 | 0.500000 | 130985 | 753/955/576 |
| 2026090907 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090907 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090907 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090907 | high | desynchronized_independent | 1.000000 | 1.000000 | 17302 | 753/955/576 |
| 2026090907 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090907 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090907 | high | shielded_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090908 | low | desynchronized_independent | 0.333333 | 0.000000 | 131029 | 884/483/389 |
| 2026090908 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090908 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090908 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090908 | mid | desynchronized_independent | 0.666667 | 0.500000 | 130985 | 884/483/389 |
| 2026090908 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090908 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090908 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090908 | high | desynchronized_independent | 1.000000 | 1.000000 | 71828 | 884/483/389 |
| 2026090908 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090908 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090908 | high | shielded_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090909 | low | desynchronized_independent | 0.333333 | 0.000000 | 131029 | 530/348/834 |
| 2026090909 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090909 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090909 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090909 | mid | desynchronized_independent | 0.666667 | 0.500000 | 130985 | 530/348/834 |
| 2026090909 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090909 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090909 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090909 | high | desynchronized_independent | 1.000000 | 1.000000 | 64881 | 530/348/834 |
| 2026090909 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090909 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090909 | high | shielded_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090910 | low | desynchronized_independent | 0.333333 | 0.000000 | 131028 | 611/131/96 |
| 2026090910 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090910 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090910 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090910 | mid | desynchronized_independent | 0.666667 | 0.505556 | 130985 | 611/131/96 |
| 2026090910 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090910 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090910 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090910 | high | desynchronized_independent | 1.000000 | 1.000000 | 130876 | 611/131/96 |

| Seed | Cap. | Method | Core frac. | On-time | Max Q (B) | Phases (ms) |
|---|---|---|---|---|---|---|
| 2026090910 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090910 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090910 | high | shielded_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090911 | low | desynchronized_independent | 0.333333 | 0.000000 | 131029 | 286/697/613 |
| 2026090911 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090911 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090911 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090911 | mid | desynchronized_independent | 0.666667 | 0.505556 | 130985 | 286/697/613 |
| 2026090911 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090911 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090911 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090911 | high | desynchronized_independent | 1.000000 | 1.000000 | 118752 | 286/697/613 |
| 2026090911 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090911 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090911 | high | shielded_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090912 | low | desynchronized_independent | 0.333333 | 0.000000 | 131028 | 72/898/820 |
| 2026090912 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090912 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090912 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090912 | mid | desynchronized_independent | 0.666667 | 0.505556 | 130985 | 72/898/820 |
| 2026090912 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090912 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090912 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090912 | high | desynchronized_independent | 1.000000 | 1.000000 | 102368 | 72/898/820 |
| 2026090912 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090912 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090912 | high | shielded_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090913 | low | desynchronized_independent | 0.333333 | 0.000000 | 131028 | 31/997/848 |
| 2026090913 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090913 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090913 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090913 | mid | desynchronized_independent | 0.666667 | 0.505556 | 130985 | 31/997/848 |
| 2026090913 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090913 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090913 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090913 | high | desynchronized_independent | 1.000000 | 1.000000 | 118490 | 31/997/848 |
| 2026090913 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090913 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090913 | high | shielded_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090914 | low | desynchronized_independent | 0.333333 | 0.000000 | 131029 | 379/396/678 |
| 2026090914 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090914 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090914 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090914 | mid | desynchronized_independent | 0.666667 | 0.500000 | 130985 | 379/396/678 |
| 2026090914 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090914 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090914 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090914 | high | desynchronized_independent | 1.000000 | 1.000000 | 126223 | 379/396/678 |
| 2026090914 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090914 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090914 | high | shielded_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090915 | low | desynchronized_independent | 0.333333 | 0.000000 | 131028 | 212/604/213 |
| 2026090915 | low | synchronized_independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090915 | low | central_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090915 | low | shielded_coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090915 | mid | desynchronized_independent | 0.666667 | 0.505556 | 130985 | 212/604/213 |
| 2026090915 | mid | synchronized_independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090915 | mid | central_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090915 | mid | shielded_coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090915 | high | desynchronized_independent | 1.000000 | 1.000000 | 130876 | 212/604/213 |
| 2026090915 | high | synchronized_independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090915 | high | central_coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |

| Seed | Cap. | Method | Core frac. | On-time | Max Q (B) | Phases (ms) |
|---|---|---|---|---|---|---|
| 2026090915 | high | shielded__coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090916 | low | desynchronized__independent | 0.333333 | 0.000000 | 131029 | 586/124/90 |
| 2026090916 | low | synchronized__independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090916 | low | central__coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090916 | low | shielded__coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090916 | mid | desynchronized__independent | 0.666667 | 0.505556 | 130985 | 586/124/90 |
| 2026090916 | mid | synchronized__independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090916 | mid | central__coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090916 | mid | shielded__coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090916 | high | desynchronized__independent | 1.000000 | 1.000000 | 130876 | 586/124/90 |
| 2026090916 | high | synchronized__independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090916 | high | central__coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090916 | high | shielded__coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090917 | low | desynchronized__independent | 0.333333 | 0.000000 | 131029 | 13/5/654 |
| 2026090917 | low | synchronized__independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090917 | low | central__coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090917 | low | shielded__coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090917 | mid | desynchronized__independent | 0.666667 | 0.500000 | 130985 | 13/5/654 |
| 2026090917 | mid | synchronized__independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090917 | mid | central__coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090917 | mid | shielded__coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090917 | high | desynchronized__independent | 1.000000 | 1.000000 | 130876 | 13/5/654 |
| 2026090917 | high | synchronized__independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090917 | high | central__coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090917 | high | shielded__coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090918 | low | desynchronized__independent | 0.333333 | 0.000000 | 131029 | 303/966/432 |
| 2026090918 | low | synchronized__independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090918 | low | central__coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090918 | low | shielded__coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090918 | mid | desynchronized__independent | 0.666667 | 0.500000 | 130985 | 303/966/432 |
| 2026090918 | mid | synchronized__independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090918 | mid | central__coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090918 | mid | shielded__coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090918 | high | desynchronized__independent | 1.000000 | 1.000000 | 91882 | 303/966/432 |
| 2026090918 | high | synchronized__independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090918 | high | central__coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090918 | high | shielded__coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090919 | low | desynchronized__independent | 0.333333 | 0.000000 | 131029 | 94/427/937 |
| 2026090919 | low | synchronized__independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090919 | low | central__coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090919 | low | shielded__coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090919 | mid | desynchronized__independent | 0.666667 | 0.500000 | 130985 | 94/427/937 |
| 2026090919 | mid | synchronized__independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090919 | mid | central__coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090919 | mid | shielded__coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090919 | high | desynchronized__independent | 1.000000 | 1.000000 | 93848 | 94/427/937 |
| 2026090919 | high | synchronized__independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090919 | high | central__coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090919 | high | shielded__coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090920 | low | desynchronized__independent | 0.333333 | 0.000000 | 131029 | 699/148/926 |
| 2026090920 | low | synchronized__independent | 0.244444 | 0.000000 | 131029 | 0/0/0 |
| 2026090920 | low | central__coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090920 | low | shielded__coordinated | 0.000000 | – | 16341 | 0/0/0 |
| 2026090920 | mid | desynchronized__independent | 0.666667 | 0.505556 | 130985 | 699/148/926 |
| 2026090920 | mid | synchronized__independent | 0.500000 | 0.000000 | 130985 | 0/0/0 |
| 2026090920 | mid | central__coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090920 | mid | shielded__coordinated | 0.333333 | 1.000000 | 130985 | 0/0/0 |
| 2026090920 | high | desynchronized__independent | 1.000000 | 1.000000 | 72483 | 699/148/926 |
| 2026090920 | high | synchronized__independent | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090920 | high | central__coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |
| 2026090920 | high | shielded__coordinated | 1.000000 | 1.000000 | 130941 | 0/0/0 |

# VI. Complete Nominal-Deadline Reconstruction

Post-release latency is result return minus actual event creation. Nominal latency is result return minus scheduled release. The 1-s quantity in the main paper is a post-release service SLO, not a complete schedule-to-result deadline.

TABLE V: Complete accepted-hardware timing reconstruction (648 retained cells; latency in ms).

| Session | Method | Cap. | Dev. | $n_c$ | Post | Nom. | Late p50 | p95 | p99 |
|---|---|---|---|---|---|---|---|---|---|
| F12_S01 | fixed_local | low | P1 | 0 | – | – | 1862.2 | 1920.9 | 1924.1 |
| F12_S01 | fixed_local | low | P2 | 0 | – | – | 760.3 | 831.8 | 834.1 |
| F12_S01 | fixed_local | low | P3 | 0 | – | – | 86.3 | 155.7 | 162.0 |
| F12_S01 | fixed_local | mid | P1 | 0 | – | – | 1828.4 | 1979.1 | 1980.6 |
| F12_S01 | fixed_local | mid | P2 | 0 | – | – | 636.8 | 775.7 | 780.4 |
| F12_S01 | fixed_local | mid | P3 | 0 | – | – | 87.5 | 165.1 | 172.5 |
| F12_S01 | fixed_local | high | P1 | 0 | – | – | 1827.5 | 1863.0 | 1913.4 |
| F12_S01 | fixed_local | high | P2 | 0 | – | – | 637.0 | 671.0 | 673.3 |
| F12_S01 | fixed_local | high | P3 | 0 | – | – | 84.4 | 156.5 | 163.5 |
| F12_S01 | fixed_summary | low | P1 | 0 | – | – | 1952.6 | 2002.0 | 2006.5 |
| F12_S01 | fixed_summary | low | P2 | 0 | – | – | 712.1 | 750.1 | 753.4 |
| F12_S01 | fixed_summary | low | P3 | 0 | – | – | 84.0 | 156.4 | 163.8 |
| F12_S01 | fixed_summary | mid | P1 | 0 | – | – | 1922.6 | 1997.1 | 2001.5 |
| F12_S01 | fixed_summary | mid | P2 | 0 | – | – | 716.0 | 804.2 | 806.5 |
| F12_S01 | fixed_summary | mid | P3 | 0 | – | – | 85.2 | 153.6 | 160.0 |
| F12_S01 | fixed_summary | high | P1 | 0 | – | – | 1925.6 | 1978.0 | 1981.4 |
| F12_S01 | fixed_summary | high | P2 | 0 | – | – | 745.5 | 802.9 | 806.8 |
| F12_S01 | fixed_summary | high | P3 | 0 | – | – | 88.3 | 159.6 | 166.3 |
| F12_S01 | fixed_full | low | P1 | 90 | 0.0000 | 0.0000 | 1887.7 | 1981.0 | 1983.3 |
| F12_S01 | fixed_full | low | P2 | 90 | 0.0000 | 0.0000 | 736.5 | 827.9 | 830.4 |
| F12_S01 | fixed_full | low | P3 | 90 | 0.0000 | 0.0000 | 79.3 | 139.5 | 145.3 |
| F12_S01 | fixed_full | mid | P1 | 90 | 0.0000 | 0.0000 | 1859.1 | 2483.1 | 2486.7 |
| F12_S01 | fixed_full | mid | P2 | 90 | 0.0000 | 0.0000 | 686.7 | 1341.8 | 1346.6 |
| F12_S01 | fixed_full | mid | P3 | 90 | 0.0000 | 0.0000 | 79.2 | 141.9 | 148.5 |
| F12_S01 | fixed_full | high | P1 | 90 | 0.9667 | 0.0000 | 2024.3 | 2101.8 | 2105.2 |
| F12_S01 | fixed_full | high | P2 | 90 | 0.9222 | 0.0000 | 830.5 | 948.8 | 952.3 |
| F12_S01 | fixed_full | high | P3 | 90 | 1.0000 | 1.0000 | 78.6 | 141.2 | 145.6 |
| F12_S01 | independent_dpp | low | P1 | 22 | 0.0000 | 0.0000 | 1851.5 | 2183.8 | 2301.2 |
| F12_S01 | independent_dpp | low | P2 | 22 | 0.0000 | 0.0000 | 661.8 | 792.4 | 1053.9 |
| F12_S01 | independent_dpp | low | P3 | 22 | 0.0000 | 0.0000 | 82.3 | 152.1 | 156.3 |
| F12_S01 | independent_dpp | mid | P1 | 45 | 0.0000 | 0.0000 | 1998.0 | 2108.8 | 2111.0 |
| F12_S01 | independent_dpp | mid | P2 | 45 | 0.0000 | 0.0000 | 754.9 | 822.0 | 824.8 |
| F12_S01 | independent_dpp | mid | P3 | 45 | 0.0000 | 0.0000 | 81.4 | 146.9 | 152.4 |
| F12_S01 | independent_dpp | high | P1 | 90 | 0.9444 | 0.0000 | 1967.9 | 2006.6 | 2009.6 |
| F12_S01 | independent_dpp | high | P2 | 90 | 0.9556 | 0.0000 | 797.6 | 836.1 | 841.8 |
| F12_S01 | independent_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 78.4 | 136.1 | 140.8 |
| F12_S01 | central_dpp | low | P1 | 0 | – | – | 2049.7 | 2151.4 | 2153.5 |
| F12_S01 | central_dpp | low | P2 | 0 | – | – | 858.4 | 942.6 | 947.1 |
| F12_S01 | central_dpp | low | P3 | 0 | – | – | 83.2 | 148.0 | 154.0 |
| F12_S01 | central_dpp | mid | P1 | 30 | 0.8000 | 0.0000 | 1910.4 | 1992.9 | 1995.4 |
| F12_S01 | central_dpp | mid | P2 | 30 | 0.8000 | 0.0000 | 732.4 | 809.6 | 813.0 |
| F12_S01 | central_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 83.1 | 152.5 | 159.3 |
| F12_S01 | central_dpp | high | P1 | 90 | 1.0000 | 0.0000 | 2070.5 | 2105.4 | 2109.1 |
| F12_S01 | central_dpp | high | P2 | 90 | 0.9889 | 0.0000 | 907.5 | 949.5 | 953.2 |
| F12_S01 | central_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 78.5 | 144.9 | 150.8 |
| F12_S01 | shielded_dpp | low | P1 | 0 | – | – | 1771.7 | 1832.4 | 1835.5 |
| F12_S01 | shielded_dpp | low | P2 | 0 | – | – | 678.1 | 749.6 | 786.5 |
| F12_S01 | shielded_dpp | low | P3 | 0 | – | – | 87.8 | 156.7 | 162.3 |
| F12_S01 | shielded_dpp | mid | P1 | 30 | 0.9000 | 0.0000 | 1934.5 | 2160.6 | 2163.7 |
| F12_S01 | shielded_dpp | mid | P2 | 30 | 1.0000 | 0.0000 | 856.4 | 964.9 | 967.9 |
| F12_S01 | shielded_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 86.3 | 149.8 | 155.9 |
| F12_S01 | shielded_dpp | high | P1 | 90 | 0.9333 | 0.0000 | 1979.2 | 2065.8 | 2070.3 |
| F12_S01 | shielded_dpp | high | P2 | 90 | 0.9444 | 0.0000 | 876.4 | 913.1 | 974.2 |
| F12_S01 | shielded_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 79.7 | 136.0 | 140.8 |
| F12_S02 | fixed_local | low | P1 | 0 | – | – | 1886.4 | 2021.6 | 2024.1 |
| F12_S02 | fixed_local | low | P2 | 0 | – | – | 594.0 | 740.2 | 743.2 |
| F12_S02 | fixed_local | low | P3 | 0 | – | – | 88.2 | 164.6 | 171.5 |
| F12_S02 | fixed_local | mid | P1 | 0 | – | – | 2003.6 | 2046.7 | 2051.4 |
| F12_S02 | fixed_local | mid | P2 | 0 | – | – | 717.1 | 761.0 | 764.1 |
| F12_S02 | fixed_local | mid | P3 | 0 | – | – | 92.2 | 161.7 | 168.4 |
| F12_S02 | fixed_local | high | P1 | 0 | – | – | 1896.5 | 1938.2 | 1941.3 |
| F12_S02 | fixed_local | high | P2 | 0 | – | – | 737.1 | 775.7 | 778.9 |
| F12_S02 | fixed_local | high | P3 | 0 | – | – | 91.3 | 160.5 | 167.6 |
| F12_S02 | fixed_summary | low | P1 | 0 | – | – | 1725.1 | 1764.9 | 1767.8 |
| F12_S02 | fixed_summary | low | P2 | 0 | – | – | 550.6 | 585.7 | 589.6 |
| F12_S02 | fixed_summary | low | P3 | 0 | – | – | 90.6 | 160.8 | 167.3 |
| F12_S02 | fixed_summary | mid | P1 | 0 | – | – | 1915.3 | 2065.5 | 2072.1 |
| F12_S02 | fixed_summary | mid | P2 | 0 | – | – | 677.3 | 1103.5 | 1106.6 |
| F12_S02 | fixed_summary | mid | P3 | 0 | – | – | 85.8 | 156.6 | 162.7 |
| F12_S02 | fixed_summary | high | P1 | 0 | – | – | 1878.3 | 1949.3 | 1959.8 |
| F12_S02 | fixed_summary | high | P2 | 0 | – | – | 713.1 | 787.0 | 798.8 |
| F12_S02 | fixed_summary | high | P3 | 0 | – | – | 89.0 | 158.6 | 165.8 |
| F12_S02 | fixed_full | low | P1 | 90 | 0.0000 | 0.0000 | 1929.0 | 2110.4 | 2114.6 |
| F12_S02 | fixed_full | low | P2 | 90 | 0.0000 | 0.0000 | 763.0 | 876.5 | 879.5 |

| Session | Method | Cap. | Dev. | $n_C$ | Post | Nom. | Late p50 | p95 | p99 |
|---|---|---|---|---|---|---|---|---|---|
| F12_S02 | fixed_full | low | P3 | 90 | 0.0000 | 0.0000 | 78.2 | 135.9 | 142.5 |
| F12_S02 | fixed_full | mid | P1 | 90 | 0.0000 | 0.0000 | 2251.6 | 2305.8 | 2309.9 |
| F12_S02 | fixed_full | mid | P2 | 90 | 0.0000 | 0.0000 | 616.5 | 733.8 | 750.5 |
| F12_S02 | fixed_full | mid | P3 | 90 | 0.0000 | 0.0000 | 76.4 | 136.6 | 142.5 |
| F12_S02 | fixed_full | high | P1 | 90 | 1.0000 | 0.0000 | 1969.5 | 1996.6 | 2001.5 |
| F12_S02 | fixed_full | high | P2 | 90 | 0.9889 | 0.0000 | 827.2 | 865.1 | 867.9 |
| F12_S02 | fixed_full | high | P3 | 90 | 1.0000 | 1.0000 | 80.0 | 145.5 | 151.7 |
| F12_S02 | independent_dpp | low | P1 | 22 | 0.0000 | 0.0000 | 1906.4 | 2060.6 | 2064.0 |
| F12_S02 | independent_dpp | low | P2 | 22 | 0.0000 | 0.0000 | 748.0 | 899.5 | 904.3 |
| F12_S02 | independent_dpp | low | P3 | 22 | 0.0000 | 0.0000 | 78.9 | 142.0 | 147.5 |
| F12_S02 | independent_dpp | mid | P1 | 45 | 0.0000 | 0.0000 | 1980.3 | 2013.9 | 2015.6 |
| F12_S02 | independent_dpp | mid | P2 | 45 | 0.0000 | 0.0000 | 703.8 | 732.1 | 734.0 |
| F12_S02 | independent_dpp | mid | P3 | 45 | 0.0000 | 0.0000 | 79.5 | 147.8 | 152.5 |
| F12_S02 | independent_dpp | high | P1 | 90 | 0.9111 | 0.0000 | 1982.0 | 2052.8 | 2056.0 |
| F12_S02 | independent_dpp | high | P2 | 90 | 0.9333 | 0.0000 | 787.2 | 831.0 | 832.9 |
| F12_S02 | independent_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 78.0 | 137.5 | 141.7 |
| F12_S02 | central_dpp | low | P1 | 0 | – | – | 2017.6 | 2154.2 | 2157.5 |
| F12_S02 | central_dpp | low | P2 | 0 | – | – | 738.5 | 861.7 | 865.2 |
| F12_S02 | central_dpp | low | P3 | 0 | – | – | 87.9 | 158.3 | 164.1 |
| F12_S02 | central_dpp | mid | P1 | 30 | 0.7333 | 0.0000 | 1936.7 | 2071.6 | 2074.8 |
| F12_S02 | central_dpp | mid | P2 | 30 | 0.9333 | 0.0000 | 881.1 | 913.6 | 916.2 |
| F12_S02 | central_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 86.2 | 150.4 | 156.4 |
| F12_S02 | central_dpp | high | P1 | 90 | 0.9222 | 0.0000 | 1942.0 | 1995.4 | 2000.0 |
| F12_S02 | central_dpp | high | P2 | 90 | 0.9444 | 0.0000 | 816.7 | 862.3 | 865.6 |
| F12_S02 | central_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 81.3 | 145.8 | 151.8 |
| F12_S02 | shielded_dpp | low | P1 | 0 | – | – | 1931.4 | 1995.7 | 1999.1 |
| F12_S02 | shielded_dpp | low | P2 | 0 | – | – | 842.4 | 903.9 | 908.0 |
| F12_S02 | shielded_dpp | low | P3 | 0 | – | – | 86.6 | 160.6 | 168.3 |
| F12_S02 | shielded_dpp | mid | P1 | 30 | 0.9000 | 0.0000 | 2120.6 | 2150.6 | 2151.7 |
| F12_S02 | shielded_dpp | mid | P2 | 30 | 0.9000 | 0.0000 | 793.9 | 826.4 | 828.9 |
| F12_S02 | shielded_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 80.3 | 148.0 | 155.9 |
| F12_S02 | shielded_dpp | high | P1 | 90 | 0.9444 | 0.0000 | 1954.9 | 2007.4 | 2039.4 |
| F12_S02 | shielded_dpp | high | P2 | 90 | 0.9333 | 0.0000 | 781.0 | 858.6 | 873.6 |
| F12_S02 | shielded_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 81.5 | 140.4 | 145.8 |
| F12_S03 | fixed_local | low | P1 | 0 | – | – | 1892.3 | 1942.8 | 1950.8 |
| F12_S03 | fixed_local | low | P2 | 0 | – | – | 676.8 | 732.8 | 744.0 |
| F12_S03 | fixed_local | low | P3 | 0 | – | – | 90.0 | 162.6 | 169.3 |
| F12_S03 | fixed_local | mid | P1 | 0 | – | – | 1823.3 | 1868.5 | 1871.1 |
| F12_S03 | fixed_local | mid | P2 | 0 | – | – | 736.6 | 837.7 | 840.2 |
| F12_S03 | fixed_local | mid | P3 | 0 | – | – | 89.2 | 160.3 | 166.4 |
| F12_S03 | fixed_local | high | P1 | 0 | – | – | 1939.5 | 2052.7 | 2061.3 |
| F12_S03 | fixed_local | high | P2 | 0 | – | – | 724.4 | 838.4 | 852.7 |
| F12_S03 | fixed_local | high | P3 | 0 | – | – | 90.8 | 167.3 | 174.3 |
| F12_S03 | fixed_summary | low | P1 | 0 | – | – | 1926.4 | 1967.6 | 1971.1 |
| F12_S03 | fixed_summary | low | P2 | 0 | – | – | 710.5 | 757.2 | 762.6 |
| F12_S03 | fixed_summary | low | P3 | 0 | – | – | 90.1 | 166.0 | 172.4 |
| F12_S03 | fixed_summary | mid | P1 | 0 | – | – | 2190.5 | 2234.0 | 2237.6 |
| F12_S03 | fixed_summary | mid | P2 | 0 | – | – | 877.1 | 918.5 | 922.6 |
| F12_S03 | fixed_summary | mid | P3 | 0 | – | – | 90.8 | 162.8 | 169.3 |
| F12_S03 | fixed_summary | high | P1 | 0 | – | – | 2024.1 | 2062.5 | 2066.0 |
| F12_S03 | fixed_summary | high | P2 | 0 | – | – | 748.8 | 787.7 | 790.7 |
| F12_S03 | fixed_summary | high | P3 | 0 | – | – | 89.3 | 160.2 | 166.3 |
| F12_S03 | fixed_full | low | P1 | 90 | 0.0000 | 0.0000 | 1963.8 | 2094.5 | 2097.1 |
| F12_S03 | fixed_full | low | P2 | 90 | 0.0000 | 0.0000 | 864.3 | 926.3 | 947.9 |
| F12_S03 | fixed_full | low | P3 | 90 | 0.0000 | 0.0000 | 78.2 | 147.4 | 153.9 |
| F12_S03 | fixed_full | mid | P1 | 90 | 0.0000 | 0.0000 | 1903.6 | 1975.4 | 1976.5 |
| F12_S03 | fixed_full | mid | P2 | 90 | 0.0000 | 0.0000 | 752.0 | 824.8 | 829.8 |
| F12_S03 | fixed_full | mid | P3 | 90 | 0.0000 | 0.0000 | 80.7 | 141.3 | 147.6 |
| F12_S03 | fixed_full | high | P1 | 90 | 0.9667 | 0.0000 | 2006.1 | 2050.0 | 2098.4 |
| F12_S03 | fixed_full | high | P2 | 90 | 0.9333 | 0.0000 | 847.8 | 892.2 | 1004.3 |
| F12_S03 | fixed_full | high | P3 | 90 | 1.0000 | 1.0000 | 86.9 | 150.9 | 156.0 |
| F12_S03 | independent_dpp | low | P1 | 22 | 0.0000 | 0.0000 | 1937.5 | 2023.7 | 2025.5 |
| F12_S03 | independent_dpp | low | P2 | 22 | 0.0000 | 0.0000 | 755.3 | 832.0 | 841.6 |
| F12_S03 | independent_dpp | low | P3 | 22 | 0.0000 | 0.0000 | 82.0 | 151.6 | 158.6 |
| F12_S03 | independent_dpp | mid | P1 | 45 | 0.0000 | 0.0000 | 7335.7 | 7390.7 | 7395.4 |
| F12_S03 | independent_dpp | mid | P2 | 45 | 0.0000 | 0.0000 | 6236.0 | 6290.3 | 6295.7 |
| F12_S03 | independent_dpp | mid | P3 | 45 | 0.0000 | 0.0000 | 4851.8 | 4918.9 | 4924.2 |
| F12_S03 | independent_dpp | high | P1 | 90 | 0.9889 | 0.0000 | 1867.0 | 2014.7 | 2018.8 |
| F12_S03 | independent_dpp | high | P2 | 90 | 0.9556 | 0.0000 | 699.5 | 814.5 | 817.0 |
| F12_S03 | independent_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 74.3 | 134.1 | 139.7 |
| F12_S03 | central_dpp | low | P1 | 0 | – | – | 1984.5 | 2024.7 | 2030.2 |
| F12_S03 | central_dpp | low | P2 | 0 | – | – | 838.4 | 875.1 | 878.3 |
| F12_S03 | central_dpp | low | P3 | 0 | – | – | 90.1 | 161.4 | 168.3 |
| F12_S03 | central_dpp | mid | P1 | 30 | 0.8667 | 0.0000 | 1949.1 | 2028.9 | 2031.1 |
| F12_S03 | central_dpp | mid | P2 | 30 | 0.7667 | 0.0000 | 915.3 | 954.2 | 957.1 |
| F12_S03 | central_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 81.0 | 147.8 | 154.9 |
| F12_S03 | central_dpp | high | P1 | 90 | 0.9667 | 0.0000 | 1963.1 | 2023.1 | 2025.9 |
| F12_S03 | central_dpp | high | P2 | 90 | 1.0000 | 0.0000 | 856.1 | 892.5 | 896.6 |
| F12_S03 | central_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 73.2 | 133.8 | 140.1 |
| F12_S03 | shielded_dpp | low | P1 | 0 | – | – | 2074.6 | 2116.7 | 2120.3 |
| F12_S03 | shielded_dpp | low | P2 | 0 | – | – | 627.9 | 665.5 | 668.1 |
| F12_S03 | shielded_dpp | low | P3 | 0 | – | – | 84.1 | 154.8 | 160.5 |
| F12_S03 | shielded_dpp | mid | P1 | 30 | 0.8000 | 0.0000 | 2003.3 | 2147.9 | 2150.1 |

| Session | Method | Cap. | Dev. | $n_c$ | Post | Nom. | Late p50 | p95 | p99 |
|---|---|---|---|---|---|---|---|---|---|
| F12_S03 | shielded_dpp | mid | P2 | 30 | 0.8333 | 0.0000 | 758.1 | 862.1 | 892.8 |
| F12_S03 | shielded_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 86.6 | 151.2 | 156.3 |
| F12_S03 | shielded_dpp | high | P1 | 90 | 0.9333 | 0.0000 | 2057.5 | 2130.4 | 2139.2 |
| F12_S03 | shielded_dpp | high | P2 | 90 | 0.9778 | 0.0000 | 939.6 | 989.1 | 992.6 |
| F12_S03 | shielded_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 77.9 | 142.0 | 146.9 |
| F12_S04 | fixed_local | low | P1 | 0 | – | – | 1879.6 | 1975.4 | 1977.9 |
| F12_S04 | fixed_local | low | P2 | 0 | – | – | 657.9 | 744.4 | 746.8 |
| F12_S04 | fixed_local | low | P3 | 0 | – | – | 91.4 | 163.7 | 170.4 |
| F12_S04 | fixed_local | mid | P1 | 0 | – | – | 1944.5 | 2017.8 | 2022.3 |
| F12_S04 | fixed_local | mid | P2 | 0 | – | – | 724.9 | 795.4 | 798.7 |
| F12_S04 | fixed_local | mid | P3 | 0 | – | – | 95.3 | 169.4 | 176.3 |
| F12_S04 | fixed_local | high | P1 | 0 | – | – | 1896.6 | 1934.6 | 1938.1 |
| F12_S04 | fixed_local | high | P2 | 0 | – | – | 743.8 | 780.2 | 783.8 |
| F12_S04 | fixed_local | high | P3 | 0 | – | – | 87.9 | 160.4 | 166.0 |
| F12_S04 | fixed_summary | low | P1 | 0 | – | – | 1958.8 | 2049.4 | 2059.2 |
| F12_S04 | fixed_summary | low | P2 | 0 | – | – | 719.0 | 800.6 | 806.0 |
| F12_S04 | fixed_summary | low | P3 | 0 | – | – | 87.6 | 156.2 | 162.2 |
| F12_S04 | fixed_summary | mid | P1 | 0 | – | – | 1929.8 | 2022.4 | 2027.0 |
| F12_S04 | fixed_summary | mid | P2 | 0 | – | – | 706.0 | 830.5 | 834.7 |
| F12_S04 | fixed_summary | mid | P3 | 0 | – | – | 89.1 | 158.3 | 164.2 |
| F12_S04 | fixed_summary | high | P1 | 0 | – | – | 1973.7 | 2015.9 | 2019.2 |
| F12_S04 | fixed_summary | high | P2 | 0 | – | – | 595.9 | 629.4 | 630.5 |
| F12_S04 | fixed_summary | high | P3 | 0 | – | – | 88.2 | 155.6 | 161.3 |
| F12_S04 | fixed_full | low | P1 | 90 | 0.0000 | 0.0000 | 2135.3 | 2299.8 | 2301.8 |
| F12_S04 | fixed_full | low | P2 | 90 | 0.0000 | 0.0000 | 843.4 | 1033.6 | 1036.2 |
| F12_S04 | fixed_full | low | P3 | 90 | 0.0000 | 0.0000 | 78.4 | 140.7 | 146.6 |
| F12_S04 | fixed_full | mid | P1 | 90 | 0.0000 | 0.0000 | 1990.7 | 2057.3 | 2060.3 |
| F12_S04 | fixed_full | mid | P2 | 90 | 0.0000 | 0.0000 | 864.7 | 897.3 | 900.8 |
| F12_S04 | fixed_full | mid | P3 | 90 | 0.0000 | 0.0000 | 78.5 | 141.6 | 146.8 |
| F12_S04 | fixed_full | high | P1 | 90 | 0.9556 | 0.0000 | 1968.2 | 2168.1 | 2172.8 |
| F12_S04 | fixed_full | high | P2 | 90 | 0.9222 | 0.0000 | 746.4 | 946.3 | 950.5 |
| F12_S04 | fixed_full | high | P3 | 90 | 1.0000 | 1.0000 | 78.6 | 142.2 | 146.8 |
| F12_S04 | independent_dpp | low | P1 | 22 | 0.0000 | 0.0000 | 2113.4 | 2196.6 | 2199.2 |
| F12_S04 | independent_dpp | low | P2 | 22 | 0.0000 | 0.0000 | 1004.2 | 1036.6 | 1038.8 |
| F12_S04 | independent_dpp | low | P3 | 22 | 0.0000 | 0.0000 | 85.1 | 154.4 | 159.8 |
| F12_S04 | independent_dpp | mid | P1 | 45 | 0.0000 | 0.0000 | 1990.7 | 2228.1 | 2232.6 |
| F12_S04 | independent_dpp | mid | P2 | 45 | 0.0000 | 0.0000 | 837.2 | 975.7 | 977.9 |
| F12_S04 | independent_dpp | mid | P3 | 45 | 0.0000 | 0.0000 | 77.9 | 143.7 | 149.7 |
| F12_S04 | independent_dpp | high | P1 | 90 | 0.9444 | 0.0000 | 2007.0 | 2091.5 | 2094.1 |
| F12_S04 | independent_dpp | high | P2 | 90 | 0.9556 | 0.0000 | 806.0 | 927.9 | 929.9 |
| F12_S04 | independent_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 80.4 | 145.8 | 151.8 |
| F12_S04 | central_dpp | low | P1 | 0 | – | – | 1985.7 | 2026.6 | 2029.4 |
| F12_S04 | central_dpp | low | P2 | 0 | – | – | 669.0 | 706.4 | 709.7 |
| F12_S04 | central_dpp | low | P3 | 0 | – | – | 85.8 | 158.2 | 163.9 |
| F12_S04 | central_dpp | mid | P1 | 30 | 0.6667 | 0.0000 | 1908.9 | 1971.8 | 1982.7 |
| F12_S04 | central_dpp | mid | P2 | 30 | 0.7667 | 0.0000 | 747.3 | 842.3 | 846.1 |
| F12_S04 | central_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 86.9 | 156.1 | 162.2 |
| F12_S04 | central_dpp | high | P1 | 90 | 0.9556 | 0.0000 | 2089.1 | 2217.9 | 2220.9 |
| F12_S04 | central_dpp | high | P2 | 90 | 0.9889 | 0.0000 | 855.7 | 922.1 | 923.9 |
| F12_S04 | central_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 74.0 | 133.8 | 139.0 |
| F12_S04 | shielded_dpp | low | P1 | 0 | – | – | 1873.0 | 1929.6 | 1934.2 |
| F12_S04 | shielded_dpp | low | P2 | 0 | – | – | 648.1 | 713.7 | 716.7 |
| F12_S04 | shielded_dpp | low | P3 | 0 | – | – | 85.0 | 156.3 | 162.7 |
| F12_S04 | shielded_dpp | mid | P1 | 30 | 0.7667 | 0.0000 | 1978.4 | 2044.0 | 2047.9 |
| F12_S04 | shielded_dpp | mid | P2 | 30 | 0.7667 | 0.0000 | 795.4 | 884.6 | 887.1 |
| F12_S04 | shielded_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 81.0 | 144.6 | 150.5 |
| F12_S04 | shielded_dpp | high | P1 | 90 | 0.9778 | 0.0000 | 1987.3 | 2025.7 | 2029.0 |
| F12_S04 | shielded_dpp | high | P2 | 90 | 0.9111 | 0.0000 | 724.4 | 854.0 | 858.5 |
| F12_S04 | shielded_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 78.4 | 144.4 | 150.5 |
| F12_S05 | fixed_local | low | P1 | 0 | – | – | 2011.8 | 2048.3 | 2050.5 |
| F12_S05 | fixed_local | low | P2 | 0 | – | – | 735.0 | 769.7 | 771.7 |
| F12_S05 | fixed_local | low | P3 | 0 | – | – | 88.0 | 158.3 | 164.5 |
| F12_S05 | fixed_local | mid | P1 | 0 | – | – | 1826.1 | 1896.2 | 1901.4 |
| F12_S05 | fixed_local | mid | P2 | 0 | – | – | 665.2 | 704.7 | 706.8 |
| F12_S05 | fixed_local | mid | P3 | 0 | – | – | 86.6 | 158.2 | 164.3 |
| F12_S05 | fixed_local | high | P1 | 0 | – | – | 1917.4 | 1963.7 | 1966.4 |
| F12_S05 | fixed_local | high | P2 | 0 | – | – | 763.8 | 800.2 | 802.8 |
| F12_S05 | fixed_local | high | P3 | 0 | – | – | 90.4 | 169.1 | 175.5 |
| F12_S05 | fixed_summary | low | P1 | 0 | – | – | 1836.5 | 1895.6 | 1899.3 |
| F12_S05 | fixed_summary | low | P2 | 0 | – | – | 624.0 | 695.4 | 698.3 |
| F12_S05 | fixed_summary | low | P3 | 0 | – | – | 88.9 | 160.6 | 166.5 |
| F12_S05 | fixed_summary | mid | P1 | 0 | – | – | 1787.3 | 1904.6 | 1908.2 |
| F12_S05 | fixed_summary | mid | P2 | 0 | – | – | 749.4 | 864.3 | 868.8 |
| F12_S05 | fixed_summary | mid | P3 | 0 | – | – | 91.0 | 163.6 | 170.1 |
| F12_S05 | fixed_summary | high | P1 | 0 | – | – | 1869.3 | 1931.6 | 1935.0 |
| F12_S05 | fixed_summary | high | P2 | 0 | – | – | 712.6 | 774.4 | 779.4 |
| F12_S05 | fixed_summary | high | P3 | 0 | – | – | 86.0 | 152.7 | 159.9 |
| F12_S05 | fixed_full | low | P1 | 90 | 0.0000 | 0.0000 | 2017.7 | 2049.2 | 2051.7 |
| F12_S05 | fixed_full | low | P2 | 90 | 0.0000 | 0.0000 | 779.2 | 840.4 | 844.0 |
| F12_S05 | fixed_full | low | P3 | 90 | 0.0000 | 0.0000 | 76.6 | 140.7 | 146.2 |
| F12_S05 | fixed_full | mid | P1 | 90 | 0.0000 | 0.0000 | 1920.6 | 2001.0 | 2049.4 |
| F12_S05 | fixed_full | mid | P2 | 90 | 0.0000 | 0.0000 | 746.0 | 797.5 | 843.3 |
| F12_S05 | fixed_full | mid | P3 | 90 | 0.0000 | 0.0000 | 78.4 | 144.6 | 151.0 |

| Session | Method | Cap. | Dev. | $n_C$ | Post | Nom. | Late p50 | p95 | p99 |
|---|---|---|---|---|---|---|---|---|---|
| F12_S05 | fixed_full | high | P1 | 90 | 0.9111 | 0.0000 | 1861.7 | 1922.9 | 1926.9 |
| F12_S05 | fixed_full | high | P2 | 90 | 0.9222 | 0.0000 | 777.4 | 829.5 | 831.5 |
| F12_S05 | fixed_full | high | P3 | 90 | 1.0000 | 1.0000 | 81.1 | 140.9 | 146.8 |
| F12_S05 | independent_dpp | low | P1 | 22 | 0.0000 | 0.0000 | 1901.3 | 1976.4 | 1981.7 |
| F12_S05 | independent_dpp | low | P2 | 22 | 0.0000 | 0.0000 | 654.1 | 712.2 | 715.8 |
| F12_S05 | independent_dpp | low | P3 | 22 | 0.0000 | 0.0000 | 81.8 | 145.1 | 151.4 |
| F12_S05 | independent_dpp | mid | P1 | 45 | 0.0000 | 0.0000 | 1841.5 | 1921.5 | 1925.4 |
| F12_S05 | independent_dpp | mid | P2 | 45 | 0.0000 | 0.0000 | 675.7 | 737.0 | 747.7 |
| F12_S05 | independent_dpp | mid | P3 | 45 | 0.0000 | 0.0000 | 74.9 | 137.5 | 142.5 |
| F12_S05 | independent_dpp | high | P1 | 90 | 0.9778 | 0.0000 | 1962.0 | 2083.4 | 2087.9 |
| F12_S05 | independent_dpp | high | P2 | 90 | 0.9778 | 0.0000 | 886.6 | 920.4 | 923.2 |
| F12_S05 | independent_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 76.8 | 139.6 | 145.5 |
| F12_S05 | central_dpp | low | P1 | 0 | – | – | 1957.7 | 1999.1 | 2002.0 |
| F12_S05 | central_dpp | low | P2 | 0 | – | – | 769.8 | 811.2 | 814.7 |
| F12_S05 | central_dpp | low | P3 | 0 | – | – | 85.3 | 153.6 | 160.1 |
| F12_S05 | central_dpp | mid | P1 | 30 | 1.0000 | 0.0000 | 1912.7 | 1940.1 | 1942.0 |
| F12_S05 | central_dpp | mid | P2 | 30 | 0.8000 | 0.0000 | 793.2 | 835.2 | 837.4 |
| F12_S05 | central_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 82.5 | 150.1 | 155.5 |
| F12_S05 | central_dpp | high | P1 | 90 | 0.9111 | 0.0000 | 1946.2 | 1994.5 | 1999.5 |
| F12_S05 | central_dpp | high | P2 | 90 | 0.9778 | 0.0000 | 792.1 | 861.7 | 865.5 |
| F12_S05 | central_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 77.4 | 141.8 | 146.7 |
| F12_S05 | shielded_dpp | low | P1 | 0 | – | – | 1893.0 | 1978.6 | 1982.3 |
| F12_S05 | shielded_dpp | low | P2 | 0 | – | – | 673.0 | 754.2 | 757.7 |
| F12_S05 | shielded_dpp | low | P3 | 0 | – | – | 84.9 | 152.9 | 159.8 |
| F12_S05 | shielded_dpp | mid | P1 | 30 | 0.5667 | 0.0000 | 1903.8 | 1973.6 | 1976.1 |
| F12_S05 | shielded_dpp | mid | P2 | 30 | 0.8333 | 0.0000 | 822.2 | 890.1 | 892.5 |
| F12_S05 | shielded_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 81.1 | 146.8 | 152.3 |
| F12_S05 | shielded_dpp | high | P1 | 90 | 0.9222 | 0.0000 | 1914.0 | 1990.0 | 1992.6 |
| F12_S05 | shielded_dpp | high | P2 | 90 | 0.9667 | 0.0000 | 701.7 | 774.2 | 778.5 |
| F12_S05 | shielded_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 80.0 | 140.8 | 147.1 |
| F12_S06 | fixed_local | low | P1 | 0 | – | – | 1932.5 | 2004.1 | 2017.5 |
| F12_S06 | fixed_local | low | P2 | 0 | – | – | 752.8 | 824.8 | 837.3 |
| F12_S06 | fixed_local | low | P3 | 0 | – | – | 90.0 | 164.6 | 171.9 |
| F12_S06 | fixed_local | mid | P1 | 0 | – | – | 1937.2 | 2042.4 | 2059.8 |
| F12_S06 | fixed_local | mid | P2 | 0 | – | – | 673.6 | 769.8 | 782.5 |
| F12_S06 | fixed_local | mid | P3 | 0 | – | – | 90.2 | 159.5 | 166.9 |
| F12_S06 | fixed_local | high | P1 | 0 | – | – | 1909.3 | 1956.4 | 1959.2 |
| F12_S06 | fixed_local | high | P2 | 0 | – | – | 785.0 | 826.4 | 829.8 |
| F12_S06 | fixed_local | high | P3 | 0 | – | – | 90.5 | 162.6 | 169.1 |
| F12_S06 | fixed_summary | low | P1 | 0 | – | – | 1785.0 | 1821.6 | 1825.8 |
| F12_S06 | fixed_summary | low | P2 | 0 | – | – | 614.8 | 652.1 | 654.9 |
| F12_S06 | fixed_summary | low | P3 | 0 | – | – | 88.8 | 157.6 | 164.1 |
| F12_S06 | fixed_summary | mid | P1 | 0 | – | – | 1889.4 | 1925.6 | 1929.1 |
| F12_S06 | fixed_summary | mid | P2 | 0 | – | – | 741.5 | 779.2 | 781.7 |
| F12_S06 | fixed_summary | mid | P3 | 0 | – | – | 85.5 | 156.6 | 163.6 |
| F12_S06 | fixed_summary | high | P1 | 0 | – | – | 1854.5 | 1988.9 | 2023.9 |
| F12_S06 | fixed_summary | high | P2 | 0 | – | – | 659.0 | 780.9 | 817.9 |
| F12_S06 | fixed_summary | high | P3 | 0 | – | – | 91.7 | 165.0 | 171.2 |
| F12_S06 | fixed_full | low | P1 | 90 | 0.0000 | 0.0000 | 2001.6 | 2059.9 | 2106.9 |
| F12_S06 | fixed_full | low | P2 | 90 | 0.0000 | 0.0000 | 694.5 | 809.4 | 850.0 |
| F12_S06 | fixed_full | low | P3 | 90 | 0.0000 | 0.0000 | 80.2 | 143.3 | 148.5 |
| F12_S06 | fixed_full | mid | P1 | 90 | 0.0000 | 0.0000 | 1890.5 | 2019.8 | 2023.1 |
| F12_S06 | fixed_full | mid | P2 | 90 | 0.0000 | 0.0000 | 652.4 | 715.4 | 735.8 |
| F12_S06 | fixed_full | mid | P3 | 90 | 0.0000 | 0.0000 | 77.1 | 141.6 | 146.7 |
| F12_S06 | fixed_full | high | P1 | 90 | 0.9667 | 0.0000 | 1966.0 | 2015.7 | 2038.6 |
| F12_S06 | fixed_full | high | P2 | 90 | 0.9111 | 0.0000 | 734.7 | 797.3 | 799.7 |
| F12_S06 | fixed_full | high | P3 | 90 | 1.0000 | 1.0000 | 80.4 | 144.3 | 150.8 |
| F12_S06 | independent_dpp | low | P1 | 22 | 0.0000 | 0.0000 | 2005.8 | 2040.9 | 2043.7 |
| F12_S06 | independent_dpp | low | P2 | 22 | 0.0000 | 0.0000 | 696.2 | 797.4 | 800.0 |
| F12_S06 | independent_dpp | low | P3 | 22 | 0.0000 | 0.0000 | 84.9 | 156.4 | 161.7 |
| F12_S06 | independent_dpp | mid | P1 | 45 | 0.0000 | 0.0000 | 1969.6 | 1996.1 | 2000.6 |
| F12_S06 | independent_dpp | mid | P2 | 45 | 0.0000 | 0.0000 | 903.6 | 932.0 | 934.8 |
| F12_S06 | independent_dpp | mid | P3 | 45 | 0.0000 | 0.0000 | 79.1 | 144.5 | 150.1 |
| F12_S06 | independent_dpp | high | P1 | 90 | 0.9222 | 0.0000 | 1912.8 | 2042.7 | 2046.9 |
| F12_S06 | independent_dpp | high | P2 | 90 | 0.9444 | 0.0000 | 730.8 | 856.4 | 859.5 |
| F12_S06 | independent_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 73.9 | 132.8 | 138.6 |
| F12_S06 | central_dpp | low | P1 | 0 | – | – | 1945.7 | 2067.9 | 2070.5 |
| F12_S06 | central_dpp | low | P2 | 0 | – | – | 746.1 | 825.0 | 827.8 |
| F12_S06 | central_dpp | low | P3 | 0 | – | – | 84.7 | 151.8 | 158.0 |
| F12_S06 | central_dpp | mid | P1 | 30 | 0.9000 | 0.0000 | 2001.6 | 2059.9 | 2070.6 |
| F12_S06 | central_dpp | mid | P2 | 30 | 0.8667 | 0.0000 | 869.7 | 935.4 | 937.7 |
| F12_S06 | central_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 89.2 | 151.7 | 158.0 |
| F12_S06 | central_dpp | high | P1 | 90 | 0.9444 | 0.0000 | 2175.3 | 2214.4 | 2217.3 |
| F12_S06 | central_dpp | high | P2 | 90 | 0.9667 | 0.0000 | 1059.9 | 1102.3 | 1105.7 |
| F12_S06 | central_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 79.4 | 154.0 | 160.2 |
| F12_S06 | shielded_dpp | low | P1 | 0 | – | – | 1933.6 | 1972.0 | 1976.6 |
| F12_S06 | shielded_dpp | low | P2 | 0 | – | – | 775.1 | 817.3 | 820.9 |
| F12_S06 | shielded_dpp | low | P3 | 0 | – | – | 89.9 | 160.3 | 166.5 |
| F12_S06 | shielded_dpp | mid | P1 | 30 | 0.8333 | 0.0000 | 1830.6 | 1960.8 | 1963.5 |
| F12_S06 | shielded_dpp | mid | P2 | 30 | 0.9000 | 0.0000 | 688.1 | 761.8 | 764.0 |
| F12_S06 | shielded_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 82.1 | 150.9 | 156.6 |
| F12_S06 | shielded_dpp | high | P1 | 90 | 0.9778 | 0.0000 | 1922.9 | 1970.4 | 1972.8 |
| F12_S06 | shielded_dpp | high | P2 | 90 | 0.9556 | 0.0000 | 698.8 | 740.1 | 744.4 |

| Session | Method | Cap. | Dev. | $n_C$ | Post | Nom. | Late p50 | p95 | p99 |
|---|---|---|---|---|---|---|---|---|---|
| F12_S06 | shielded_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 88.2 | 149.6 | 155.1 |
| F12_S07 | fixed_local | low | P1 | 0 | – | – | 1944.9 | 2011.2 | 2014.7 |
| F12_S07 | fixed_local | low | P2 | 0 | – | – | 730.1 | 809.6 | 812.8 |
| F12_S07 | fixed_local | low | P3 | 0 | – | – | 88.8 | 160.6 | 167.3 |
| F12_S07 | fixed_local | mid | P1 | 0 | – | – | 1868.9 | 1943.8 | 1948.2 |
| F12_S07 | fixed_local | mid | P2 | 0 | – | – | 629.0 | 699.1 | 702.5 |
| F12_S07 | fixed_local | mid | P3 | 0 | – | – | 93.9 | 164.7 | 171.4 |
| F12_S07 | fixed_local | high | P1 | 0 | – | – | 2100.5 | 2245.8 | 2249.1 |
| F12_S07 | fixed_local | high | P2 | 0 | – | – | 895.1 | 1020.1 | 1024.0 |
| F12_S07 | fixed_local | high | P3 | 0 | – | – | 88.5 | 159.7 | 166.7 |
| F12_S07 | fixed_summary | low | P1 | 0 | – | – | 1997.9 | 2132.3 | 2136.3 |
| F12_S07 | fixed_summary | low | P2 | 0 | – | – | 675.1 | 801.3 | 804.5 |
| F12_S07 | fixed_summary | low | P3 | 0 | – | – | 84.1 | 151.1 | 156.9 |
| F12_S07 | fixed_summary | mid | P1 | 0 | – | – | 1927.0 | 1978.8 | 1985.2 |
| F12_S07 | fixed_summary | mid | P2 | 0 | – | – | 612.5 | 680.9 | 684.0 |
| F12_S07 | fixed_summary | mid | P3 | 0 | – | – | 87.5 | 157.8 | 165.0 |
| F12_S07 | fixed_summary | high | P1 | 0 | – | – | 2181.8 | 2223.8 | 2227.5 |
| F12_S07 | fixed_summary | high | P2 | 0 | – | – | 613.1 | 751.0 | 755.5 |
| F12_S07 | fixed_summary | high | P3 | 0 | – | – | 84.2 | 154.1 | 161.1 |
| F12_S07 | fixed_full | low | P1 | 90 | 0.0000 | 0.0000 | 1825.8 | 1945.2 | 1947.3 |
| F12_S07 | fixed_full | low | P2 | 90 | 0.0000 | 0.0000 | 557.2 | 701.7 | 705.8 |
| F12_S07 | fixed_full | low | P3 | 90 | 0.0000 | 0.0000 | 81.1 | 143.6 | 149.2 |
| F12_S07 | fixed_full | mid | P1 | 90 | 0.0000 | 0.0000 | 2023.6 | 2274.3 | 2277.8 |
| F12_S07 | fixed_full | mid | P2 | 90 | 0.0000 | 0.0000 | 771.6 | 952.8 | 955.2 |
| F12_S07 | fixed_full | mid | P3 | 90 | 0.0000 | 0.0000 | 77.4 | 139.8 | 145.4 |
| F12_S07 | fixed_full | high | P1 | 90 | 0.9778 | 0.0000 | 1983.4 | 2025.4 | 2028.5 |
| F12_S07 | fixed_full | high | P2 | 90 | 0.9333 | 0.0000 | 793.0 | 850.9 | 872.5 |
| F12_S07 | fixed_full | high | P3 | 90 | 1.0000 | 1.0000 | 79.3 | 145.9 | 150.9 |
| F12_S07 | independent_dpp | low | P1 | 22 | 0.0000 | 0.0000 | 1909.6 | 2101.8 | 2103.8 |
| F12_S07 | independent_dpp | low | P2 | 22 | 0.0000 | 0.0000 | 717.8 | 874.3 | 876.8 |
| F12_S07 | independent_dpp | low | P3 | 22 | 0.0000 | 0.0000 | 83.9 | 149.3 | 155.4 |
| F12_S07 | independent_dpp | mid | P1 | 45 | 0.0000 | 0.0000 | 2049.3 | 2104.3 | 2107.5 |
| F12_S07 | independent_dpp | mid | P2 | 45 | 0.0000 | 0.0000 | 941.2 | 966.4 | 970.3 |
| F12_S07 | independent_dpp | mid | P3 | 45 | 0.0000 | 0.0000 | 85.9 | 151.4 | 156.4 |
| F12_S07 | independent_dpp | high | P1 | 90 | 0.9667 | 0.0000 | 1967.2 | 2170.5 | 2176.7 |
| F12_S07 | independent_dpp | high | P2 | 90 | 0.8889 | 0.0000 | 782.6 | 931.8 | 1045.7 |
| F12_S07 | independent_dpp | high | P3 | 90 | 0.9889 | 0.9889 | 81.2 | 328.2 | 333.1 |
| F12_S07 | central_dpp | low | P1 | 0 | – | – | 2036.6 | 2077.1 | 2082.3 |
| F12_S07 | central_dpp | low | P2 | 0 | – | – | 723.5 | 764.1 | 768.0 |
| F12_S07 | central_dpp | low | P3 | 0 | – | – | 88.3 | 156.0 | 161.8 |
| F12_S07 | central_dpp | mid | P1 | 30 | 0.8667 | 0.0000 | 2137.4 | 2178.3 | 2188.0 |
| F12_S07 | central_dpp | mid | P2 | 30 | 0.8667 | 0.0000 | 861.1 | 927.6 | 931.2 |
| F12_S07 | central_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 87.2 | 159.0 | 164.3 |
| F12_S07 | central_dpp | high | P1 | 90 | 0.9778 | 0.0000 | 2061.6 | 2089.0 | 2093.5 |
| F12_S07 | central_dpp | high | P2 | 90 | 0.9333 | 0.0000 | 864.5 | 941.0 | 944.8 |
| F12_S07 | central_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 81.4 | 144.8 | 150.1 |
| F12_S07 | shielded_dpp | low | P1 | 0 | – | – | 1894.6 | 1993.4 | 1996.3 |
| F12_S07 | shielded_dpp | low | P2 | 0 | – | – | 628.4 | 734.7 | 738.1 |
| F12_S07 | shielded_dpp | low | P3 | 0 | – | – | 85.6 | 158.3 | 163.7 |
| F12_S07 | shielded_dpp | mid | P1 | 30 | 0.9333 | 0.0000 | 2068.2 | 2134.6 | 2137.8 |
| F12_S07 | shielded_dpp | mid | P2 | 30 | 0.9333 | 0.0000 | 716.9 | 752.4 | 755.3 |
| F12_S07 | shielded_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 79.0 | 149.3 | 154.3 |
| F12_S07 | shielded_dpp | high | P1 | 90 | 0.9889 | 0.0000 | 1891.0 | 1956.7 | 1959.9 |
| F12_S07 | shielded_dpp | high | P2 | 90 | 0.9556 | 0.0000 | 704.9 | 781.2 | 833.6 |
| F12_S07 | shielded_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 79.4 | 144.9 | 151.5 |
| F12_S08 | fixed_local | low | P1 | 0 | – | – | 1957.4 | 2013.6 | 2016.1 |
| F12_S08 | fixed_local | low | P2 | 0 | – | – | 685.9 | 732.8 | 735.9 |
| F12_S08 | fixed_local | low | P3 | 0 | – | – | 87.0 | 158.9 | 165.4 |
| F12_S08 | fixed_local | mid | P1 | 0 | – | – | 1817.2 | 1937.7 | 1954.1 |
| F12_S08 | fixed_local | mid | P2 | 0 | – | – | 614.7 | 724.4 | 744.3 |
| F12_S08 | fixed_local | mid | P3 | 0 | – | – | 88.0 | 158.5 | 164.9 |
| F12_S08 | fixed_local | high | P1 | 0 | – | – | 1937.6 | 2017.9 | 2035.3 |
| F12_S08 | fixed_local | high | P2 | 0 | – | – | 683.8 | 776.3 | 784.7 |
| F12_S08 | fixed_local | high | P3 | 0 | – | – | 89.9 | 164.2 | 170.0 |
| F12_S08 | fixed_summary | low | P1 | 0 | – | – | 1999.1 | 2079.8 | 2083.7 |
| F12_S08 | fixed_summary | low | P2 | 0 | – | – | 735.8 | 778.9 | 786.1 |
| F12_S08 | fixed_summary | low | P3 | 0 | – | – | 88.0 | 158.4 | 164.5 |
| F12_S08 | fixed_summary | mid | P1 | 0 | – | – | 1992.5 | 2077.9 | 2082.7 |
| F12_S08 | fixed_summary | mid | P2 | 0 | – | – | 745.7 | 848.4 | 854.3 |
| F12_S08 | fixed_summary | mid | P3 | 0 | – | – | 89.3 | 158.2 | 164.7 |
| F12_S08 | fixed_summary | high | P1 | 0 | – | – | 2038.2 | 2116.8 | 2119.5 |
| F12_S08 | fixed_summary | high | P2 | 0 | – | – | 750.9 | 807.1 | 811.8 |
| F12_S08 | fixed_summary | high | P3 | 0 | – | – | 87.3 | 161.2 | 167.8 |
| F12_S08 | fixed_full | low | P1 | 90 | 0.0000 | 0.0000 | 2158.3 | 2203.4 | 2206.0 |
| F12_S08 | fixed_full | low | P2 | 90 | 0.0000 | 0.0000 | 978.1 | 1030.7 | 1034.2 |
| F12_S08 | fixed_full | low | P3 | 90 | 0.0000 | 0.0000 | 75.1 | 136.5 | 142.3 |
| F12_S08 | fixed_full | mid | P1 | 90 | 0.0000 | 0.0000 | 1956.9 | 2041.8 | 2044.2 |
| F12_S08 | fixed_full | mid | P2 | 90 | 0.0000 | 0.0000 | 904.9 | 1014.7 | 1018.5 |
| F12_S08 | fixed_full | mid | P3 | 90 | 0.0000 | 0.0000 | 80.4 | 142.8 | 147.6 |
| F12_S08 | fixed_full | high | P1 | 90 | 0.9333 | 0.0000 | 2053.0 | 2190.7 | 2193.1 |
| F12_S08 | fixed_full | high | P2 | 90 | 0.9556 | 0.0000 | 908.0 | 1033.1 | 1036.6 |
| F12_S08 | fixed_full | high | P3 | 90 | 1.0000 | 1.0000 | 80.5 | 142.0 | 146.8 |
| F12_S08 | independent_dpp | low | P1 | 22 | 0.0000 | 0.0000 | 1874.7 | 2010.2 | 2012.1 |

| Session | Method | Cap. | Dev. | $n_C$ | Post | Nom. | Late p50 | p95 | p99 |
|---|---|---|---|---|---|---|---|---|---|
| F12_S08 | independent_dpp | low | P2 | 22 | 0.0000 | 0.0000 | 773.6 | 889.6 | 892.9 |
| F12_S08 | independent_dpp | low | P3 | 22 | 0.0000 | 0.0000 | 82.8 | 150.6 | 158.3 |
| F12_S08 | independent_dpp | mid | P1 | 45 | 0.0000 | 0.0000 | 1854.8 | 1911.5 | 1913.6 |
| F12_S08 | independent_dpp | mid | P2 | 45 | 0.0000 | 0.0000 | 664.3 | 721.6 | 724.3 |
| F12_S08 | independent_dpp | mid | P3 | 45 | 0.0000 | 0.0000 | 78.2 | 140.8 | 145.7 |
| F12_S08 | independent_dpp | high | P1 | 90 | 0.9222 | 0.0000 | 2146.9 | 2229.6 | 2231.4 |
| F12_S08 | independent_dpp | high | P2 | 90 | 0.9333 | 0.0000 | 995.3 | 1104.8 | 1107.4 |
| F12_S08 | independent_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 75.3 | 141.2 | 146.0 |
| F12_S08 | central_dpp | low | P1 | 0 | – | – | 2023.5 | 2362.2 | 2365.6 |
| F12_S08 | central_dpp | low | P2 | 0 | – | – | 750.6 | 1083.7 | 1087.1 |
| F12_S08 | central_dpp | low | P3 | 0 | – | – | 83.0 | 154.3 | 161.6 |
| F12_S08 | central_dpp | mid | P1 | 30 | 0.9667 | 0.0000 | 2057.4 | 2105.0 | 2107.2 |
| F12_S08 | central_dpp | mid | P2 | 30 | 0.6667 | 0.0000 | 784.2 | 856.1 | 858.2 |
| F12_S08 | central_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 79.3 | 146.0 | 152.4 |
| F12_S08 | central_dpp | high | P1 | 90 | 0.9222 | 0.0000 | 2048.7 | 2112.0 | 2115.0 |
| F12_S08 | central_dpp | high | P2 | 90 | 0.9556 | 0.0000 | 814.7 | 873.6 | 877.0 |
| F12_S08 | central_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 80.4 | 143.1 | 148.8 |
| F12_S08 | shielded_dpp | low | P1 | 0 | – | – | 1979.4 | 2072.9 | 2075.3 |
| F12_S08 | shielded_dpp | low | P2 | 0 | – | – | 746.4 | 815.6 | 819.0 |
| F12_S08 | shielded_dpp | low | P3 | 0 | – | – | 87.2 | 154.6 | 160.7 |
| F12_S08 | shielded_dpp | mid | P1 | 30 | 0.7333 | 0.0000 | 1895.4 | 1977.8 | 1980.2 |
| F12_S08 | shielded_dpp | mid | P2 | 30 | 0.7000 | 0.0000 | 754.9 | 834.4 | 837.4 |
| F12_S08 | shielded_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 84.4 | 149.1 | 155.6 |
| F12_S08 | shielded_dpp | high | P1 | 90 | 0.9556 | 0.0000 | 2070.2 | 2145.7 | 2150.4 |
| F12_S08 | shielded_dpp | high | P2 | 90 | 0.9556 | 0.0000 | 946.3 | 1008.9 | 1011.7 |
| F12_S08 | shielded_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 72.2 | 135.7 | 140.7 |
| F12_S09 | fixed_local | low | P1 | 0 | – | – | 2062.2 | 2148.9 | 2153.1 |
| F12_S09 | fixed_local | low | P2 | 0 | – | – | 835.3 | 908.6 | 912.9 |
| F12_S09 | fixed_local | low | P3 | 0 | – | – | 91.8 | 166.5 | 174.5 |
| F12_S09 | fixed_local | mid | P1 | 0 | – | – | 2017.6 | 2068.9 | 2073.1 |
| F12_S09 | fixed_local | mid | P2 | 0 | – | – | 734.4 | 788.0 | 793.2 |
| F12_S09 | fixed_local | mid | P3 | 0 | – | – | 84.3 | 154.2 | 160.5 |
| F12_S09 | fixed_local | high | P1 | 0 | – | – | 2105.4 | 2162.6 | 2166.2 |
| F12_S09 | fixed_local | high | P2 | 0 | – | – | 862.7 | 933.5 | 937.1 |
| F12_S09 | fixed_local | high | P3 | 0 | – | – | 90.6 | 162.1 | 167.7 |
| F12_S09 | fixed_summary | low | P1 | 0 | – | – | 2035.6 | 2133.6 | 2137.1 |
| F12_S09 | fixed_summary | low | P2 | 0 | – | – | 740.9 | 828.5 | 832.8 |
| F12_S09 | fixed_summary | low | P3 | 0 | – | – | 91.4 | 166.4 | 172.2 |
| F12_S09 | fixed_summary | mid | P1 | 0 | – | – | 2067.9 | 2207.0 | 2209.7 |
| F12_S09 | fixed_summary | mid | P2 | 0 | – | – | 731.5 | 865.4 | 869.0 |
| F12_S09 | fixed_summary | mid | P3 | 0 | – | – | 89.9 | 163.7 | 169.8 |
| F12_S09 | fixed_summary | high | P1 | 0 | – | – | 1931.7 | 2076.8 | 2079.4 |
| F12_S09 | fixed_summary | high | P2 | 0 | – | – | 656.3 | 775.4 | 778.8 |
| F12_S09 | fixed_summary | high | P3 | 0 | – | – | 84.7 | 152.8 | 158.7 |
| F12_S09 | fixed_full | low | P1 | 90 | 0.0000 | 0.0000 | 2058.8 | 2117.3 | 2124.4 |
| F12_S09 | fixed_full | low | P2 | 90 | 0.0000 | 0.0000 | 756.7 | 835.5 | 843.0 |
| F12_S09 | fixed_full | low | P3 | 90 | 0.0000 | 0.0000 | 77.2 | 141.2 | 148.3 |
| F12_S09 | fixed_full | mid | P1 | 90 | 0.0000 | 0.0000 | 2005.3 | 2062.5 | 2089.6 |
| F12_S09 | fixed_full | mid | P2 | 90 | 0.0000 | 0.0000 | 877.8 | 931.8 | 935.3 |
| F12_S09 | fixed_full | mid | P3 | 90 | 0.0000 | 0.0000 | 77.9 | 136.7 | 142.7 |
| F12_S09 | fixed_full | high | P1 | 90 | 0.9111 | 0.0000 | 2041.2 | 2101.3 | 2106.6 |
| F12_S09 | fixed_full | high | P2 | 90 | 0.9556 | 0.0000 | 795.5 | 870.5 | 874.5 |
| F12_S09 | fixed_full | high | P3 | 90 | 1.0000 | 1.0000 | 77.1 | 141.8 | 147.0 |
| F12_S09 | independent_dpp | low | P1 | 22 | 0.0000 | 0.0000 | 2044.8 | 2140.4 | 2142.8 |
| F12_S09 | independent_dpp | low | P2 | 22 | 0.0000 | 0.0000 | 886.4 | 937.7 | 959.9 |
| F12_S09 | independent_dpp | low | P3 | 22 | 0.0000 | 0.0000 | 85.2 | 148.8 | 157.8 |
| F12_S09 | independent_dpp | mid | P1 | 45 | 0.0000 | 0.0000 | 2163.0 | 2195.9 | 2197.6 |
| F12_S09 | independent_dpp | mid | P2 | 45 | 0.0000 | 0.0000 | 1015.9 | 1045.1 | 1065.2 |
| F12_S09 | independent_dpp | mid | P3 | 45 | 0.0000 | 0.0000 | 78.9 | 140.5 | 145.3 |
| F12_S09 | independent_dpp | high | P1 | 90 | 0.9333 | 0.0000 | 2125.5 | 2172.7 | 2174.9 |
| F12_S09 | independent_dpp | high | P2 | 90 | 0.9333 | 0.0000 | 851.2 | 955.2 | 959.5 |
| F12_S09 | independent_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 74.2 | 135.0 | 139.5 |
| F12_S09 | central_dpp | low | P1 | 0 | – | – | 1974.6 | 2062.7 | 2073.2 |
| F12_S09 | central_dpp | low | P2 | 0 | – | – | 700.8 | 778.4 | 783.7 |
| F12_S09 | central_dpp | low | P3 | 0 | – | – | 88.6 | 160.8 | 166.0 |
| F12_S09 | central_dpp | mid | P1 | 30 | 0.9000 | 0.0000 | 2105.6 | 2140.7 | 2143.5 |
| F12_S09 | central_dpp | mid | P2 | 30 | 0.7667 | 0.0000 | 957.1 | 996.2 | 1018.1 |
| F12_S09 | central_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 85.9 | 153.4 | 158.5 |
| F12_S09 | central_dpp | high | P1 | 90 | 0.9556 | 0.0000 | 2328.7 | 2368.8 | 2373.6 |
| F12_S09 | central_dpp | high | P2 | 90 | 0.9778 | 0.0000 | 997.1 | 1036.6 | 1039.8 |
| F12_S09 | central_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 68.1 | 135.1 | 142.1 |
| F12_S09 | shielded_dpp | low | P1 | 0 | – | – | 1968.5 | 2040.6 | 2046.2 |
| F12_S09 | shielded_dpp | low | P2 | 0 | – | – | 790.4 | 868.6 | 870.5 |
| F12_S09 | shielded_dpp | low | P3 | 0 | – | – | 92.7 | 162.5 | 168.8 |
| F12_S09 | shielded_dpp | mid | P1 | 30 | 0.8667 | 0.0000 | 2063.5 | 2130.1 | 2140.9 |
| F12_S09 | shielded_dpp | mid | P2 | 30 | 0.9000 | 0.0000 | 834.2 | 885.0 | 888.3 |
| F12_S09 | shielded_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 80.4 | 145.5 | 150.6 |
| F12_S09 | shielded_dpp | high | P1 | 90 | 0.9444 | 0.0000 | 2087.2 | 2138.8 | 2140.9 |
| F12_S09 | shielded_dpp | high | P2 | 90 | 0.9444 | 0.0000 | 824.1 | 897.0 | 899.7 |
| F12_S09 | shielded_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 73.9 | 137.8 | 143.0 |
| F12_S10 | fixed_local | low | P1 | 0 | – | – | 1963.5 | 2064.8 | 2068.2 |
| F12_S10 | fixed_local | low | P2 | 0 | – | – | 761.3 | 837.0 | 839.8 |
| F12_S10 | fixed_local | low | P3 | 0 | – | – | 91.5 | 166.4 | 173.2 |

| Session | Method | Cap. | Dev. | $n_C$ | Post | Nom. | Late p50 | p95 | p99 |
|---|---|---|---|---|---|---|---|---|---|
| F12_S10 | fixed_local | mid | P1 | 0 | – | – | 2011.0 | 2050.3 | 2053.7 |
| F12_S10 | fixed_local | mid | P2 | 0 | – | – | 715.1 | 755.4 | 758.7 |
| F12_S10 | fixed_local | mid | P3 | 0 | – | – | 90.9 | 162.7 | 169.3 |
| F12_S10 | fixed_local | high | P1 | 0 | – | – | 1937.4 | 2000.6 | 2003.6 |
| F12_S10 | fixed_local | high | P2 | 0 | – | – | 668.6 | 723.4 | 726.7 |
| F12_S10 | fixed_local | high | P3 | 0 | – | – | 93.9 | 163.6 | 170.2 |
| F12_S10 | fixed_summary | low | P1 | 0 | – | – | 2013.5 | 2103.8 | 2108.3 |
| F12_S10 | fixed_summary | low | P2 | 0 | – | – | 823.8 | 913.5 | 917.0 |
| F12_S10 | fixed_summary | low | P3 | 0 | – | – | 84.6 | 153.6 | 160.5 |
| F12_S10 | fixed_summary | mid | P1 | 0 | – | – | 2057.6 | 2136.9 | 2140.5 |
| F12_S10 | fixed_summary | mid | P2 | 0 | – | – | 946.8 | 1014.3 | 1018.1 |
| F12_S10 | fixed_summary | mid | P3 | 0 | – | – | 84.4 | 155.9 | 163.6 |
| F12_S10 | fixed_summary | high | P1 | 0 | – | – | 1943.4 | 2129.7 | 2134.3 |
| F12_S10 | fixed_summary | high | P2 | 0 | – | – | 764.8 | 939.7 | 943.1 |
| F12_S10 | fixed_summary | high | P3 | 0 | – | – | 87.3 | 163.1 | 170.0 |
| F12_S10 | fixed_full | low | P1 | 90 | 0.0000 | 0.0000 | 1967.7 | 2037.9 | 2041.3 |
| F12_S10 | fixed_full | low | P2 | 90 | 0.0000 | 0.0000 | 858.5 | 951.5 | 953.8 |
| F12_S10 | fixed_full | low | P3 | 90 | 0.0000 | 0.0000 | 76.8 | 138.7 | 143.3 |
| F12_S10 | fixed_full | mid | P1 | 90 | 0.0000 | 0.0000 | 2146.3 | 2221.4 | 2223.5 |
| F12_S10 | fixed_full | mid | P2 | 90 | 0.0000 | 0.0000 | 847.3 | 890.1 | 894.4 |
| F12_S10 | fixed_full | mid | P3 | 90 | 0.0000 | 0.0000 | 74.8 | 132.3 | 136.9 |
| F12_S10 | fixed_full | high | P1 | 90 | 0.9778 | 0.0000 | 2165.6 | 2199.9 | 2204.4 |
| F12_S10 | fixed_full | high | P2 | 90 | 0.9667 | 0.0000 | 871.9 | 912.1 | 916.5 |
| F12_S10 | fixed_full | high | P3 | 90 | 1.0000 | 1.0000 | 81.3 | 142.9 | 147.6 |
| F12_S10 | independent_dpp | low | P1 | 22 | 0.0000 | 0.0000 | 1921.7 | 2011.9 | 2016.7 |
| F12_S10 | independent_dpp | low | P2 | 22 | 0.0000 | 0.0000 | 653.8 | 788.2 | 791.3 |
| F12_S10 | independent_dpp | low | P3 | 22 | 0.0000 | 0.0000 | 82.9 | 150.7 | 156.6 |
| F12_S10 | independent_dpp | mid | P1 | 45 | 0.0000 | 0.0000 | 1887.2 | 1966.4 | 1968.3 |
| F12_S10 | independent_dpp | mid | P2 | 45 | 0.0000 | 0.0000 | 773.0 | 838.0 | 840.0 |
| F12_S10 | independent_dpp | mid | P3 | 45 | 0.0000 | 0.0000 | 75.8 | 141.7 | 147.3 |
| F12_S10 | independent_dpp | high | P1 | 90 | 0.8556 | 0.0000 | 1965.1 | 2090.0 | 2093.2 |
| F12_S10 | independent_dpp | high | P2 | 90 | 0.9667 | 0.0000 | 882.8 | 981.7 | 984.1 |
| F12_S10 | independent_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 76.0 | 142.0 | 147.0 |
| F12_S10 | central_dpp | low | P1 | 0 | – | – | 2041.7 | 2115.9 | 2119.5 |
| F12_S10 | central_dpp | low | P2 | 0 | – | – | 834.2 | 905.3 | 909.7 |
| F12_S10 | central_dpp | low | P3 | 0 | – | – | 89.3 | 164.3 | 169.0 |
| F12_S10 | central_dpp | mid | P1 | 30 | 0.9667 | 0.0000 | 2076.9 | 2105.3 | 2151.2 |
| F12_S10 | central_dpp | mid | P2 | 30 | 0.9000 | 0.0000 | 850.4 | 929.6 | 931.8 |
| F12_S10 | central_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 80.3 | 148.5 | 154.8 |
| F12_S10 | central_dpp | high | P1 | 90 | 0.9556 | 0.0000 | 2040.7 | 2102.1 | 2106.1 |
| F12_S10 | central_dpp | high | P2 | 90 | 0.9667 | 0.0000 | 905.9 | 940.3 | 944.7 |
| F12_S10 | central_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 76.4 | 140.4 | 146.7 |
| F12_S10 | shielded_dpp | low | P1 | 0 | – | – | 2085.7 | 2128.5 | 2130.9 |
| F12_S10 | shielded_dpp | low | P2 | 0 | – | – | 938.2 | 977.5 | 981.4 |
| F12_S10 | shielded_dpp | low | P3 | 0 | – | – | 89.8 | 160.9 | 166.9 |
| F12_S10 | shielded_dpp | mid | P1 | 30 | 0.8000 | 0.0000 | 1980.5 | 2053.7 | 2058.1 |
| F12_S10 | shielded_dpp | mid | P2 | 30 | 0.7667 | 0.0000 | 729.0 | 827.8 | 830.2 |
| F12_S10 | shielded_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 86.9 | 154.8 | 160.6 |
| F12_S10 | shielded_dpp | high | P1 | 90 | 0.9222 | 0.0000 | 2020.4 | 2096.1 | 2123.6 |
| F12_S10 | shielded_dpp | high | P2 | 90 | 0.9778 | 0.0000 | 879.9 | 928.6 | 932.5 |
| F12_S10 | shielded_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 80.1 | 143.5 | 149.0 |
| F12_S11 | fixed_local | low | P1 | 0 | – | – | 1923.5 | 1986.9 | 2001.2 |
| F12_S11 | fixed_local | low | P2 | 0 | – | – | 743.9 | 820.1 | 835.6 |
| F12_S11 | fixed_local | low | P3 | 0 | – | – | 89.1 | 161.4 | 168.1 |
| F12_S11 | fixed_local | mid | P1 | 0 | – | – | 2032.4 | 2109.7 | 2112.1 |
| F12_S11 | fixed_local | mid | P2 | 0 | – | – | 748.3 | 824.6 | 828.8 |
| F12_S11 | fixed_local | mid | P3 | 0 | – | – | 93.9 | 169.5 | 176.2 |
| F12_S11 | fixed_local | high | P1 | 0 | – | – | 2031.0 | 2105.3 | 2109.9 |
| F12_S11 | fixed_local | high | P2 | 0 | – | – | 736.1 | 814.3 | 818.7 |
| F12_S11 | fixed_local | high | P3 | 0 | – | – | 90.1 | 162.4 | 168.3 |
| F12_S11 | fixed_summary | low | P1 | 0 | – | – | 1928.4 | 1983.9 | 1990.1 |
| F12_S11 | fixed_summary | low | P2 | 0 | – | – | 639.4 | 714.9 | 725.7 |
| F12_S11 | fixed_summary | low | P3 | 0 | – | – | 82.4 | 153.2 | 159.2 |
| F12_S11 | fixed_summary | mid | P1 | 0 | – | – | 2087.9 | 2136.3 | 2140.6 |
| F12_S11 | fixed_summary | mid | P2 | 0 | – | – | 824.7 | 887.7 | 894.9 |
| F12_S11 | fixed_summary | mid | P3 | 0 | – | – | 86.8 | 157.3 | 162.5 |
| F12_S11 | fixed_summary | high | P1 | 0 | – | – | 1941.6 | 1992.7 | 1995.2 |
| F12_S11 | fixed_summary | high | P2 | 0 | – | – | 648.9 | 703.8 | 707.1 |
| F12_S11 | fixed_summary | high | P3 | 0 | – | – | 86.9 | 155.8 | 163.5 |
| F12_S11 | fixed_full | low | P1 | 90 | 0.0000 | 0.0000 | 1942.0 | 2002.2 | 2006.4 |
| F12_S11 | fixed_full | low | P2 | 90 | 0.0000 | 0.0000 | 724.1 | 806.1 | 808.5 |
| F12_S11 | fixed_full | low | P3 | 90 | 0.0000 | 0.0000 | 85.2 | 150.6 | 155.4 |
| F12_S11 | fixed_full | mid | P1 | 90 | 0.0000 | 0.0000 | 2070.3 | 2149.2 | 2152.6 |
| F12_S11 | fixed_full | mid | P2 | 90 | 0.0000 | 0.0000 | 938.8 | 980.4 | 983.0 |
| F12_S11 | fixed_full | mid | P3 | 90 | 0.0000 | 0.0000 | 79.7 | 142.6 | 148.2 |
| F12_S11 | fixed_full | high | P1 | 90 | 0.9111 | 0.0000 | 2113.8 | 2201.4 | 2203.3 |
| F12_S11 | fixed_full | high | P2 | 90 | 0.9333 | 0.0000 | 912.5 | 1038.8 | 1042.4 |
| F12_S11 | fixed_full | high | P3 | 90 | 1.0000 | 1.0000 | 76.8 | 135.8 | 141.6 |
| F12_S11 | independent_dpp | low | P1 | 22 | 0.0000 | 0.0000 | 1986.8 | 2019.7 | 2022.3 |
| F12_S11 | independent_dpp | low | P2 | 22 | 0.0000 | 0.0000 | 764.0 | 802.3 | 804.8 |
| F12_S11 | independent_dpp | low | P3 | 22 | 0.0000 | 0.0000 | 88.9 | 156.0 | 162.7 |
| F12_S11 | independent_dpp | mid | P1 | 45 | 0.0000 | 0.0000 | 2001.6 | 2087.2 | 2091.6 |
| F12_S11 | independent_dpp | mid | P2 | 45 | 0.0000 | 0.0000 | 824.6 | 904.2 | 906.9 |

| Session | Method | Cap. | Dev. | $n_c$ | Post | Nom. | Late p50 | p95 | p99 |
|---|---|---|---|---|---|---|---|---|---|
| F12_S11 | independent_dpp | mid | P3 | 45 | 0.0000 | 0.0000 | 78.9 | 141.5 | 146.5 |
| F12_S11 | independent_dpp | high | P1 | 90 | 0.9000 | 0.0000 | 2133.6 | 2182.2 | 2184.9 |
| F12_S11 | independent_dpp | high | P2 | 90 | 0.9444 | 0.0000 | 981.2 | 1046.9 | 1051.6 |
| F12_S11 | independent_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 71.8 | 129.6 | 136.5 |
| F12_S11 | central_dpp | low | P1 | 0 | – | – | 2043.7 | 2116.0 | 2120.4 |
| F12_S11 | central_dpp | low | P2 | 0 | – | – | 783.2 | 865.3 | 868.8 |
| F12_S11 | central_dpp | low | P3 | 0 | – | – | 93.5 | 162.4 | 168.8 |
| F12_S11 | central_dpp | mid | P1 | 30 | 0.8000 | 0.0000 | 2106.5 | 2167.1 | 2169.4 |
| F12_S11 | central_dpp | mid | P2 | 30 | 0.8333 | 0.0000 | 934.4 | 991.7 | 995.4 |
| F12_S11 | central_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 80.0 | 146.7 | 151.1 |
| F12_S11 | central_dpp | high | P1 | 90 | 0.8889 | 0.0000 | 1923.1 | 1979.1 | 1982.6 |
| F12_S11 | central_dpp | high | P2 | 90 | 0.9444 | 0.0000 | 689.4 | 782.6 | 785.7 |
| F12_S11 | central_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 75.7 | 135.1 | 140.8 |
| F12_S11 | shielded_dpp | low | P1 | 0 | – | – | 2014.5 | 2077.7 | 2080.4 |
| F12_S11 | shielded_dpp | low | P2 | 0 | – | – | 735.4 | 818.5 | 822.9 |
| F12_S11 | shielded_dpp | low | P3 | 0 | – | – | 87.8 | 158.6 | 165.2 |
| F12_S11 | shielded_dpp | mid | P1 | 30 | 0.8333 | 0.0000 | 1884.2 | 1947.6 | 1950.1 |
| F12_S11 | shielded_dpp | mid | P2 | 30 | 0.8667 | 0.0000 | 802.2 | 834.0 | 882.2 |
| F12_S11 | shielded_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 79.7 | 142.8 | 149.4 |
| F12_S11 | shielded_dpp | high | P1 | 90 | 0.9000 | 0.0000 | 1866.9 | 1926.8 | 1928.9 |
| F12_S11 | shielded_dpp | high | P2 | 90 | 0.9333 | 0.0000 | 651.9 | 721.0 | 724.2 |
| F12_S11 | shielded_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 78.9 | 143.3 | 149.7 |
| F12_S12 | fixed_local | low | P1 | 0 | – | – | 2021.6 | 2060.8 | 2064.9 |
| F12_S12 | fixed_local | low | P2 | 0 | – | – | 851.3 | 891.4 | 896.0 |
| F12_S12 | fixed_local | low | P3 | 0 | – | – | 89.4 | 162.6 | 169.8 |
| F12_S12 | fixed_local | mid | P1 | 0 | – | – | 2065.4 | 2110.6 | 2113.4 |
| F12_S12 | fixed_local | mid | P2 | 0 | – | – | 880.8 | 918.2 | 921.2 |
| F12_S12 | fixed_local | mid | P3 | 0 | – | – | 95.8 | 167.4 | 173.3 |
| F12_S12 | fixed_local | high | P1 | 0 | – | – | 2028.6 | 2158.6 | 2162.3 |
| F12_S12 | fixed_local | high | P2 | 0 | – | – | 775.1 | 837.5 | 843.8 |
| F12_S12 | fixed_local | high | P3 | 0 | – | – | 92.3 | 164.3 | 171.5 |
| F12_S12 | fixed_summary | low | P1 | 0 | – | – | 1910.3 | 1985.2 | 1989.2 |
| F12_S12 | fixed_summary | low | P2 | 0 | – | – | 718.9 | 806.6 | 808.8 |
| F12_S12 | fixed_summary | low | P3 | 0 | – | – | 88.1 | 159.9 | 165.1 |
| F12_S12 | fixed_summary | mid | P1 | 0 | – | – | 1893.9 | 1959.5 | 1962.4 |
| F12_S12 | fixed_summary | mid | P2 | 0 | – | – | 613.9 | 675.0 | 678.6 |
| F12_S12 | fixed_summary | mid | P3 | 0 | – | – | 88.1 | 155.1 | 161.6 |
| F12_S12 | fixed_summary | high | P1 | 0 | – | – | 2217.8 | 2256.9 | 2260.5 |
| F12_S12 | fixed_summary | high | P2 | 0 | – | – | 1057.1 | 1098.5 | 1102.7 |
| F12_S12 | fixed_summary | high | P3 | 0 | – | – | 82.4 | 150.6 | 156.6 |
| F12_S12 | fixed_full | low | P1 | 90 | 0.0000 | 0.0000 | 2056.0 | 2127.3 | 2128.7 |
| F12_S12 | fixed_full | low | P2 | 90 | 0.0000 | 0.0000 | 831.5 | 884.5 | 889.1 |
| F12_S12 | fixed_full | low | P3 | 90 | 0.0000 | 0.0000 | 85.1 | 157.7 | 163.5 |
| F12_S12 | fixed_full | mid | P1 | 90 | 0.0000 | 0.0000 | 1859.9 | 1997.8 | 2002.3 |
| F12_S12 | fixed_full | mid | P2 | 90 | 0.0000 | 0.0000 | 697.3 | 841.4 | 844.7 |
| F12_S12 | fixed_full | mid | P3 | 90 | 0.0000 | 0.0000 | 83.0 | 145.3 | 151.3 |
| F12_S12 | fixed_full | high | P1 | 90 | 0.9222 | 0.0000 | 2070.3 | 2135.0 | 2139.0 |
| F12_S12 | fixed_full | high | P2 | 90 | 0.9444 | 0.0000 | 956.1 | 1001.6 | 1003.5 |
| F12_S12 | fixed_full | high | P3 | 90 | 1.0000 | 1.0000 | 80.0 | 138.6 | 145.2 |
| F12_S12 | independent_dpp | low | P1 | 22 | 0.0000 | 0.0000 | 1001.7 | 1038.7 | 1041.5 |
| F12_S12 | independent_dpp | low | P2 | 22 | 0.0000 | 0.0000 | 864.1 | 915.7 | 918.1 |
| F12_S12 | independent_dpp | low | P3 | 22 | 0.0000 | 0.0000 | 85.0 | 157.5 | 162.6 |
| F12_S12 | independent_dpp | mid | P1 | 45 | 0.0000 | 0.0000 | 2161.5 | 2193.0 | 2197.7 |
| F12_S12 | independent_dpp | mid | P2 | 45 | 0.0000 | 0.0000 | 860.3 | 896.0 | 897.8 |
| F12_S12 | independent_dpp | mid | P3 | 45 | 0.0000 | 0.0000 | 77.0 | 138.3 | 143.6 |
| F12_S12 | independent_dpp | high | P1 | 90 | 0.9444 | 0.0000 | 2004.9 | 2052.8 | 2056.1 |
| F12_S12 | independent_dpp | high | P2 | 90 | 0.9444 | 0.0000 | 778.7 | 819.1 | 822.4 |
| F12_S12 | independent_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 74.8 | 135.8 | 141.5 |
| F12_S12 | central_dpp | low | P1 | 0 | – | – | 2072.6 | 2369.1 | 2372.7 |
| F12_S12 | central_dpp | low | P2 | 0 | – | – | 865.3 | 1186.7 | 1191.2 |
| F12_S12 | central_dpp | low | P3 | 0 | – | – | 86.1 | 161.1 | 167.6 |
| F12_S12 | central_dpp | mid | P1 | 30 | 0.5333 | 0.0000 | 1883.9 | 1935.5 | 1939.8 |
| F12_S12 | central_dpp | mid | P2 | 30 | 0.7667 | 0.0000 | 659.2 | 732.8 | 735.7 |
| F12_S12 | central_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 83.3 | 144.6 | 150.7 |
| F12_S12 | central_dpp | high | P1 | 90 | 0.9222 | 0.0000 | 2095.8 | 2187.1 | 2190.5 |
| F12_S12 | central_dpp | high | P2 | 90 | 0.9444 | 0.0000 | 846.9 | 919.8 | 922.5 |
| F12_S12 | central_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 81.8 | 147.6 | 152.4 |
| F12_S12 | shielded_dpp | low | P1 | 0 | – | – | 1879.5 | 1960.7 | 1964.6 |
| F12_S12 | shielded_dpp | low | P2 | 0 | – | – | 695.3 | 756.2 | 759.7 |
| F12_S12 | shielded_dpp | low | P3 | 0 | – | – | 87.3 | 156.0 | 161.3 |
| F12_S12 | shielded_dpp | mid | P1 | 30 | 0.9000 | 0.0000 | 2041.9 | 2073.2 | 2075.5 |
| F12_S12 | shielded_dpp | mid | P2 | 30 | 0.9333 | 0.0000 | 803.7 | 835.2 | 837.5 |
| F12_S12 | shielded_dpp | mid | P3 | 30 | 1.0000 | 1.0000 | 83.1 | 147.1 | 152.7 |
| F12_S12 | shielded_dpp | high | P1 | 90 | 0.9444 | 0.0000 | 1962.8 | 2023.5 | 2026.7 |
| F12_S12 | shielded_dpp | high | P2 | 90 | 0.9444 | 0.0000 | 736.5 | 774.2 | 776.1 |
| F12_S12 | shielded_dpp | high | P3 | 90 | 1.0000 | 1.0000 | 74.7 | 136.9 | 142.8 |

## VII. Release-Lateness Diagnostic

Independent nominal clocks, the three-probe barrier, unequal first-iteration waiting, relative sleep, and no absolute catch-up are source/log supported. The specific OS/network reason for the recurring proposal-arrival order is UNKNOWN.

TABLE VI: Release-lateness diagnostic from accepted formal timestamps (ms).

| Device | Events | p50 | p95 | p99 | Unproven residual |
|---|---|---|---|---|---|
| P1 | 19440 | 1972.34 | 2173.36 | 2326.13 | The exact OS/network reason that proposals repeatedly reached the barrier in P3/P2/P1 order, and residual per-session variation, are UNKNOWN. |
| P2 | 19440 | 763.90 | 976.78 | 1080.30 | The exact OS/network reason that proposals repeatedly reached the barrier in P3/P2/P1 order, and residual per-session variation, are UNKNOWN. |
| P3 | 19440 | 83.56 | 154.08 | 167.94 | The exact OS/network reason that proposals repeatedly reached the barrier in P3/P2/P1 order, and residual per-session variation, are UNKNOWN. |

## VIII. Finite-n Scaling Conformance

This is discrete-event conformance, not a 32-probe hardware scalability experiment.

TABLE VII: All finite-$n$ discrete-event conformance cells.

| $n$ | $\rho$ | $C$ (B) | $k^*$ | Sim. $k$ | Theory frac. | Sim. frac. | Match |
|---|---|---|---|---|---|---|---|
| 3 | 1/3 | 131072 | 0 | 0 | 0.000000 | 0.000000 | PASS |
| 3 | 1/2 | 196608 | 1 | 1 | 0.333333 | 0.333333 | PASS |
| 3 | 2/3 | 262144 | 1 | 1 | 0.333333 | 0.333333 | PASS |
| 3 | 3/4 | 294912 | 2 | 2 | 0.666667 | 0.666667 | PASS |
| 3 | 1/1 | 393216 | 3 | 3 | 1.000000 | 1.000000 | PASS |
| 8 | 1/3 | 349525 | 1 | 1 | 0.125000 | 0.125000 | PASS |
| 8 | 1/2 | 524288 | 3 | 3 | 0.375000 | 0.375000 | PASS |
| 8 | 2/3 | 699050 | 4 | 4 | 0.500000 | 0.500000 | PASS |
| 8 | 3/4 | 786432 | 5 | 5 | 0.625000 | 0.625000 | PASS |
| 8 | 1/1 | 1048576 | 8 | 8 | 1.000000 | 1.000000 | PASS |
| 16 | 1/3 | 699050 | 3 | 3 | 0.187500 | 0.187500 | PASS |
| 16 | 1/2 | 1048576 | 6 | 6 | 0.375000 | 0.375000 | PASS |
| 16 | 2/3 | 1398101 | 9 | 9 | 0.562500 | 0.562500 | PASS |
| 16 | 3/4 | 1572864 | 11 | 11 | 0.687500 | 0.687500 | PASS |
| 16 | 1/1 | 2097152 | 16 | 16 | 1.000000 | 1.000000 | PASS |
| 32 | 1/3 | 1398101 | 7 | 7 | 0.218750 | 0.218750 | PASS |
| 32 | 1/2 | 2097152 | 13 | 13 | 0.406250 | 0.406250 | PASS |
| 32 | 2/3 | 2796202 | 19 | 19 | 0.593750 | 0.593750 | PASS |
| 32 | 3/4 | 3145728 | 22 | 22 | 0.687500 | 0.687500 | PASS |
| 32 | 1/1 | 4194304 | 32 | 32 | 1.000000 | 1.000000 | PASS |

## IX. Accepted and Excluded Attempts

Accepted attempts are S01-a01, S02-a01, S03-a02, S04-a02, S05-a01, S06-a01, S07-a02, and S08–S12-a01. S03-a01, S04-a01, and S07-a01 are excluded infrastructure attempts and are not stitched into accepted sessions.

## X. Detector and Replay Limitations

The workload is frozen CICIoT2023-derived feature-vector replay, not packet replay or an uncontrolled deployment. The formal occurrence set excludes Recon-PortScan, and exact feature-vector overlap exists across frozen partitions. Perfect correctness on the formal occurrences is therefore a controlled diagnostic rather than an IDS-superiority result.

## XI. Raw Audit and Analysis Validation

The full-raw audit rebuilt 216/216 accepted arms and 58,320 measured events with zero raw-to-metric mismatch and zero shared-cap violation across 63,720 service rounds. Its verdict is PASS_WITH_ISSUES because the raw bundle lacks direct artifacts for the S03-a01 SSD incident and the S07 pre-hardware launcher abort. The supplementary analyses pass validation with 432 parameter-sweep rows, 240 desynchronization rows, 648 timing-reconstruction rows, and 20 scaling rows, with zero retained failures.

## XII. Artifact Access

Machine-readable CSV/JSON sources, number provenance, claim audit, build report, validation report, and SHA-256 manifest accompany this supplementary PDF. Third-party dataset redistribution rights remain outside the scope of this package.